

03 - Minimax

Pattern Recognition and Machine Learning, Lab 3

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Test next week

Test: 16 October

- At the **beginning of the lab**
- Topics: **Bayesian** and **non-Bayesian** tasks
- Theory and worked problems
- Also things from the **assignments**, e.g. what a Python function computes
- Bring a **calculator**
- No phone, laptop or notes

Bonus tasks

- Handed in **in person** after the lab
- You **explain** the task and convince us that the solution is correct



Cannot come to the test? Let me know in advance.

Recap

Recap: Bayesian decision making

? What do we need to know to find the Bayesian strategy?

$$R(q) = \sum_{x \in X} \sum_{k \in K} p_{XK}(x, k) W(k, q(x)), \quad q^* = \arg \min_{q: X \rightarrow D} R(q)$$

We need

- $p(x | k)$: likelihoods
- $p(k)$: **priors**
- $W(k, d)$: the **loss**

With the 0/1 loss

$$\begin{aligned} q^*(x) &= \arg \max_k p(k | x) \\ &= \arg \max_k p(x | k) p(k) \end{aligned}$$

The prior is inside the posterior.

Recap: the reject option

? When is it better to answer "I do not know"?

Decisions $D = K \cup \{\text{not known}\}$ and the loss

$$W(k, d) = \begin{cases} 0 & \text{if } d = k \\ 1 & \text{if } d \neq k \text{ and } d \neq \text{not known} \\ \varepsilon & \text{if } d = \text{not known} \end{cases}$$

The Bayesian strategy becomes

$$q^*(x) = \begin{cases} \arg \max_k p(k | x) & \text{if } 1 - \max_k p(k | x) < \varepsilon \\ \text{not known} & \text{otherwise} \end{cases}$$

When Bayes is not enough

Priors are unknown

A plate reader used **anywhere** in the country: every crossroad has different $p(k)$.

The loss is hard to set

What is the price of an overlooked tumour compared with a false alarm?

A guarantee is required

"At most 1 % of dangerous states may be overlooked."



What can we still do when we know only the likelihoods $p(x | k)$?

Non-Bayesian tasks

Error on class k : $\epsilon(k) = \sum_{x: q(x) \neq k} p(x | k)$

Neyman-Pearson

$$\min_q \epsilon(2)$$

subject to $\epsilon(1) \leq \bar{\epsilon}$

1 dangerous, 2 normal: few false alarms,
overlooked danger bounded

Wald

$$\min_{q: X \rightarrow D} \max_{k \in \{1, 2\}} \kappa(k)$$

subject to

$$\epsilon(1) \leq \epsilon, \epsilon(2) \leq \epsilon$$

decisions $D = K \cup \{?\}$, the answer "?"
is no error

$$\text{undecided: } \kappa(k) = \sum_{x: q(x)=?} p(x | k)$$

Minimax

$$\min_q \max_{k \in K} \epsilon(k)$$

the worst class is as good as possible

today

! For two classes, all three are solved by **thresholding the likelihood ratio** $r(x) = \frac{p(x|1)}{p(x|2)}$

Discrete minimax

The minimax task

$$q^* = \arg \min_{q: X \rightarrow K} \max_{k \in K} \epsilon(k), \quad \epsilon(k) = \sum_{x: q(x) \neq k} p(x | k)$$

? What is $\epsilon(k)$ in words? What do we need, and what is missing compared with Bayes?

For two classes: decide by the likelihood ratio

$$r(x) = \frac{p(x | 1)}{p(x | 2)}, \quad q(x) = \begin{cases} 1 & \text{if } r(x) > \lambda \\ 2 & \text{if } r(x) \leq \lambda \end{cases}$$

One number gives the whole strategy: the threshold λ .

Lecture: the inverse ratio $p(x | 2)/p(x | 1)$ with the threshold μ . Assignment: this form.

Discrete example: female or male?

Observation x = (height, weight), $K = \{1, 2\} = \{F, M\}$ (female, male)

$p(x \mid F)$

$p(x \mid M)$

	u-light	light	avg	heavy		u-light	light	avg	heavy
short	.197	.145	.094	.017	short	.011	.005	.011	.011
normal	.077	.299	.145	.017	normal	.005	.071	.408	.038
tall	.001	.008	.000	.000	tall	.002	.014	.255	.169

? 1. What is $r(x) = \frac{p(x|M)}{p(x|F)}$ for each cell?

? 2. How do we find the minimax strategy q^* ?

? 3. What are its errors $\epsilon(F)$ and $\epsilon(M)$?

Derivation: the likelihood ratio and its order

$$r(x) = p(x \mid M) / p(x \mid F)$$

	u-light	light	avg	heavy
short	0.056	0.034	0.117	0.647
normal	0.065	0.237	2.814	2.235
tall	2.000	1.750	∞	∞

Index i after sorting by r (from 0)

	u-light	light	avg	heavy
short	1	0	3	5
normal	2	4	9	8
tall	7	6	10	11

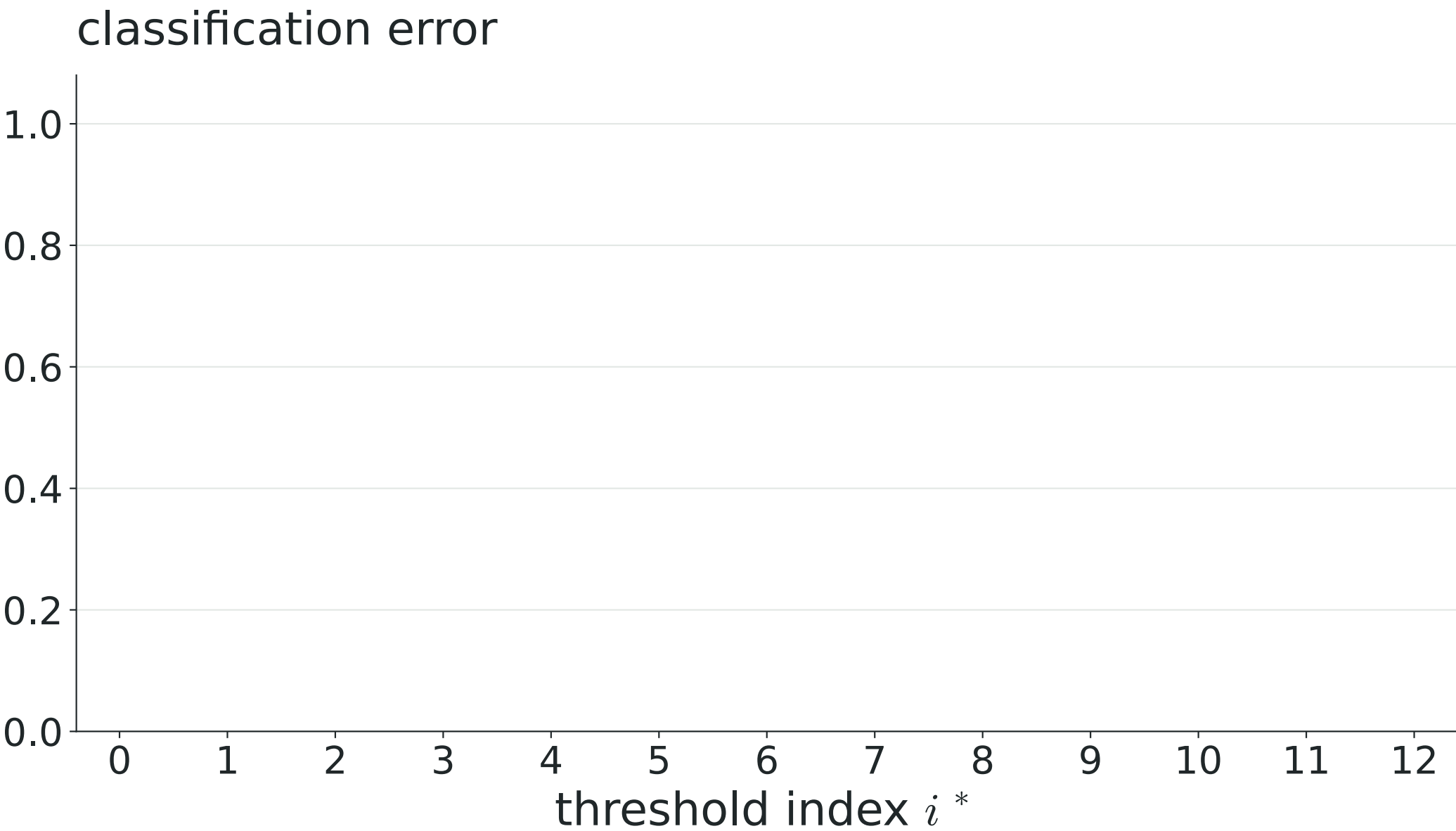
Every threshold on r is one index $i^* \in \{0, 1, \dots, 12\}$:

$$q(x_i) = \begin{cases} M & \text{if } i \geq i^* \\ F & \text{if } i < i^* \end{cases}$$

$$\epsilon(F) = \sum_{i \geq i^*} p(x_i \mid F), \qquad \epsilon(M) = \sum_{i < i^*} p(x_i \mid M)$$

Derivation: errors for every threshold

threshold i^*	0	1	2	3	4	5	6	7	8	9	10	11	12
M if $r \geq$	.034	.056	.065	.117	.237	.647	1.75	2.00	2.24	2.81	∞	∞	never
$\epsilon(F)$													
$\epsilon(M)$													
max													



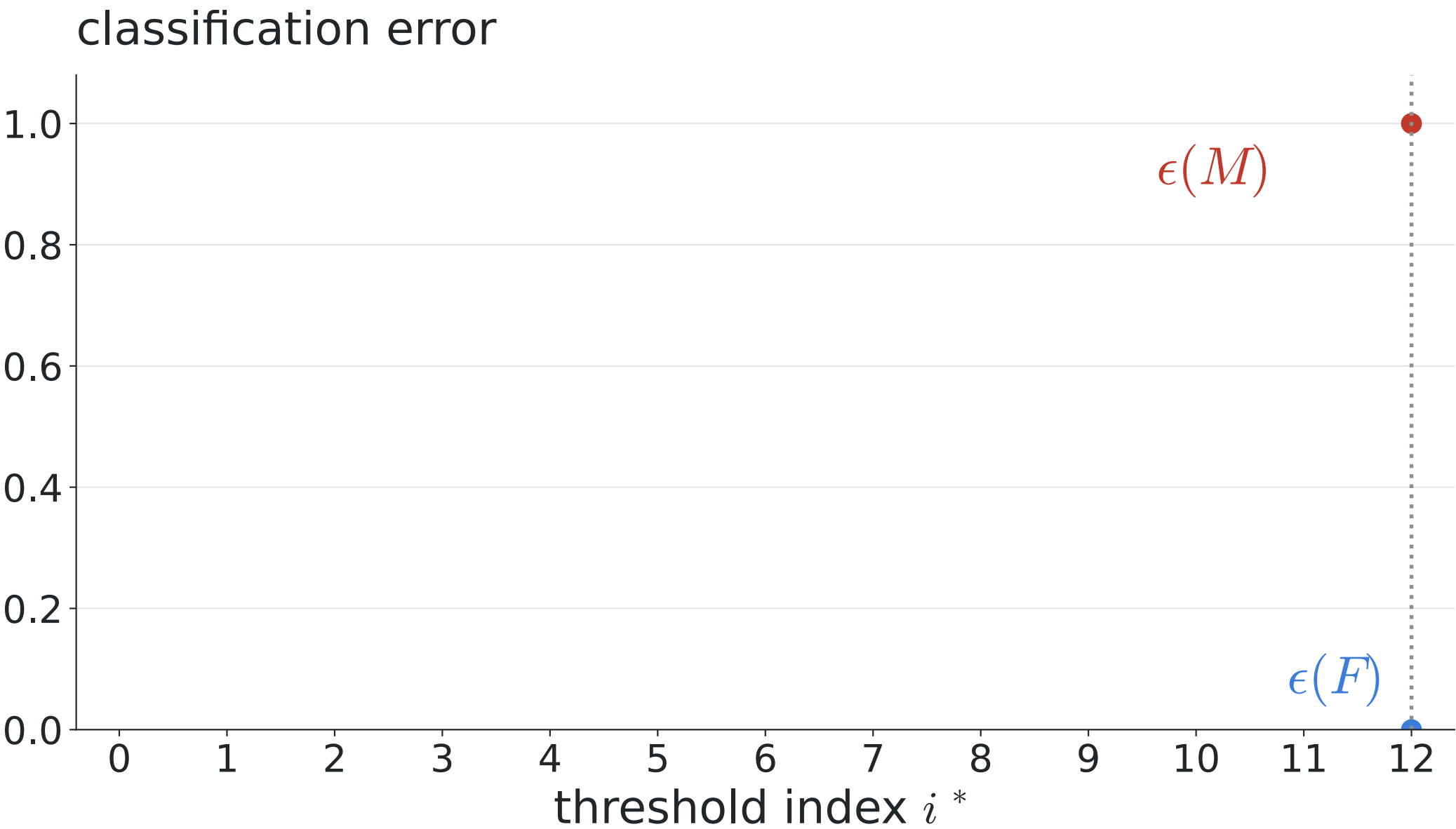
Sorted by r , decide M for $i \geq i^*$:

$$\epsilon(F) = \sum_{i \geq i^*} p(x_i \mid F)$$

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$\epsilon(F)$													0
$\epsilon(M)$													1
max													1



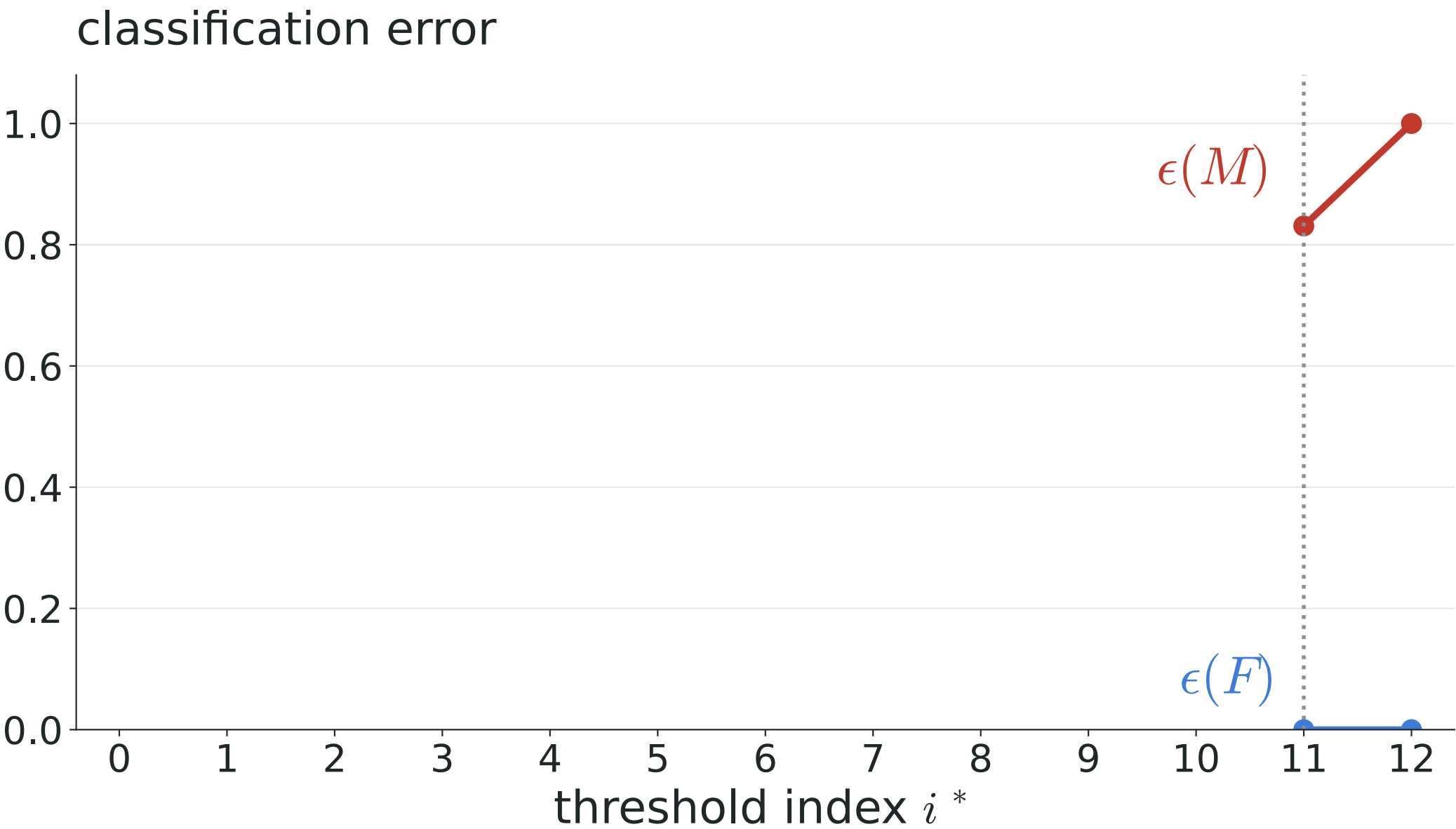
Sorted by r , decide M for $i \geq i^*$:

$$\epsilon(F) = \sum_{i \geq i^*} p(x_i | F)$$
$$\epsilon(M) = \sum_{i < i^*} p(x_i | M)$$

$i^* = 12$: nothing is decided M , i.e. we always decide F .

Derivation: errors for every threshold

threshold i^*	0	1	2	3	4	5	6	7	8	9	10	11	12
M if $r \geq$	.034	.056	.065	.117	.237	.647	1.75	2.00	2.24	2.81	∞	∞	never
$\epsilon(F)$												0	0
$\epsilon(M)$												.831	1
max												.831	1



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$\epsilon(F)$											0	0	0
$\epsilon(M)$											.576	.831	1
max											.576	.831	1



Sorted by r , decide M for $i \geq i^*$:

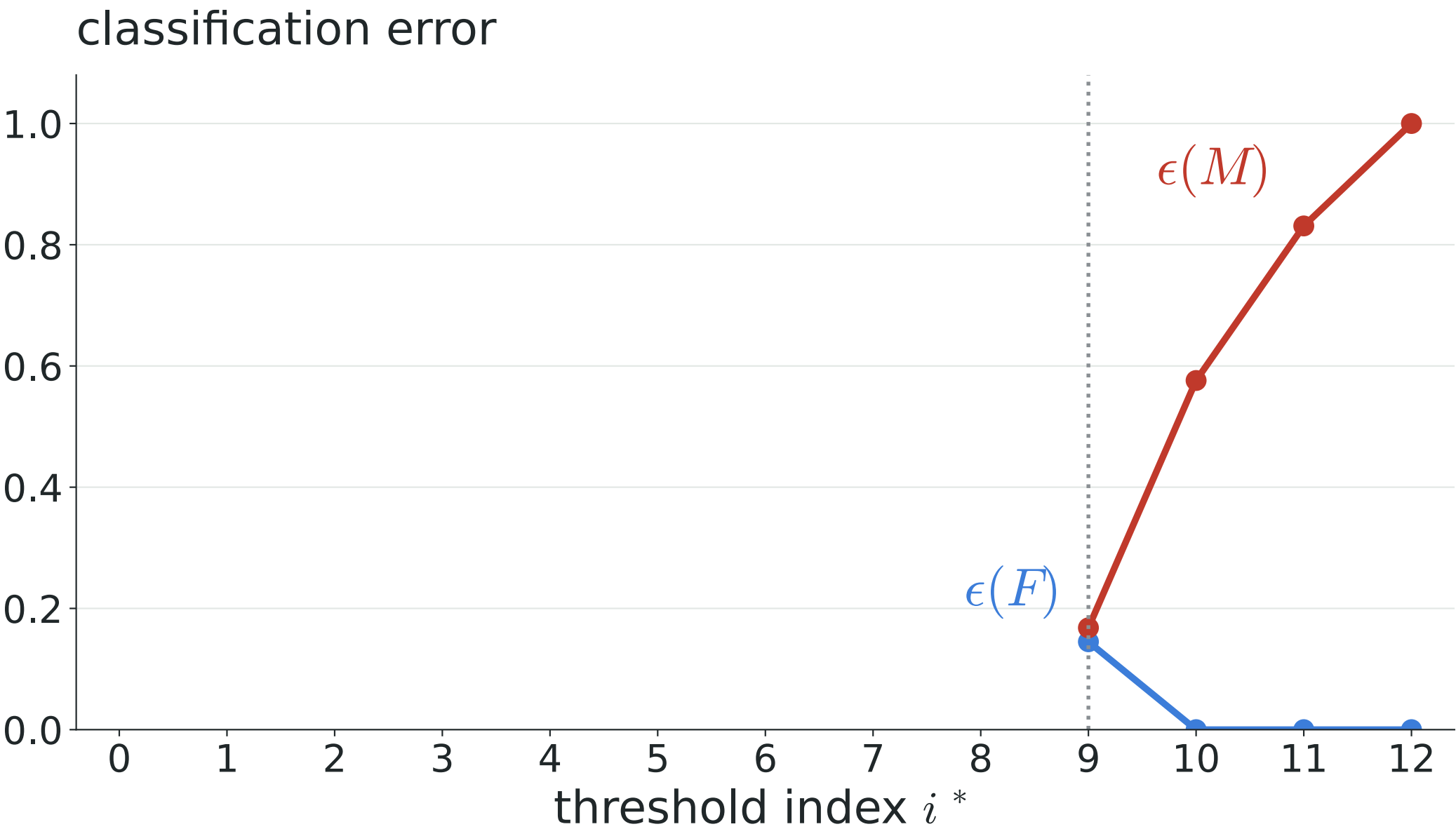
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$\epsilon(F)$										.145	0	0	0
$\epsilon(M)$										.168	.576	.831	1
max										.168	.576	.831	1



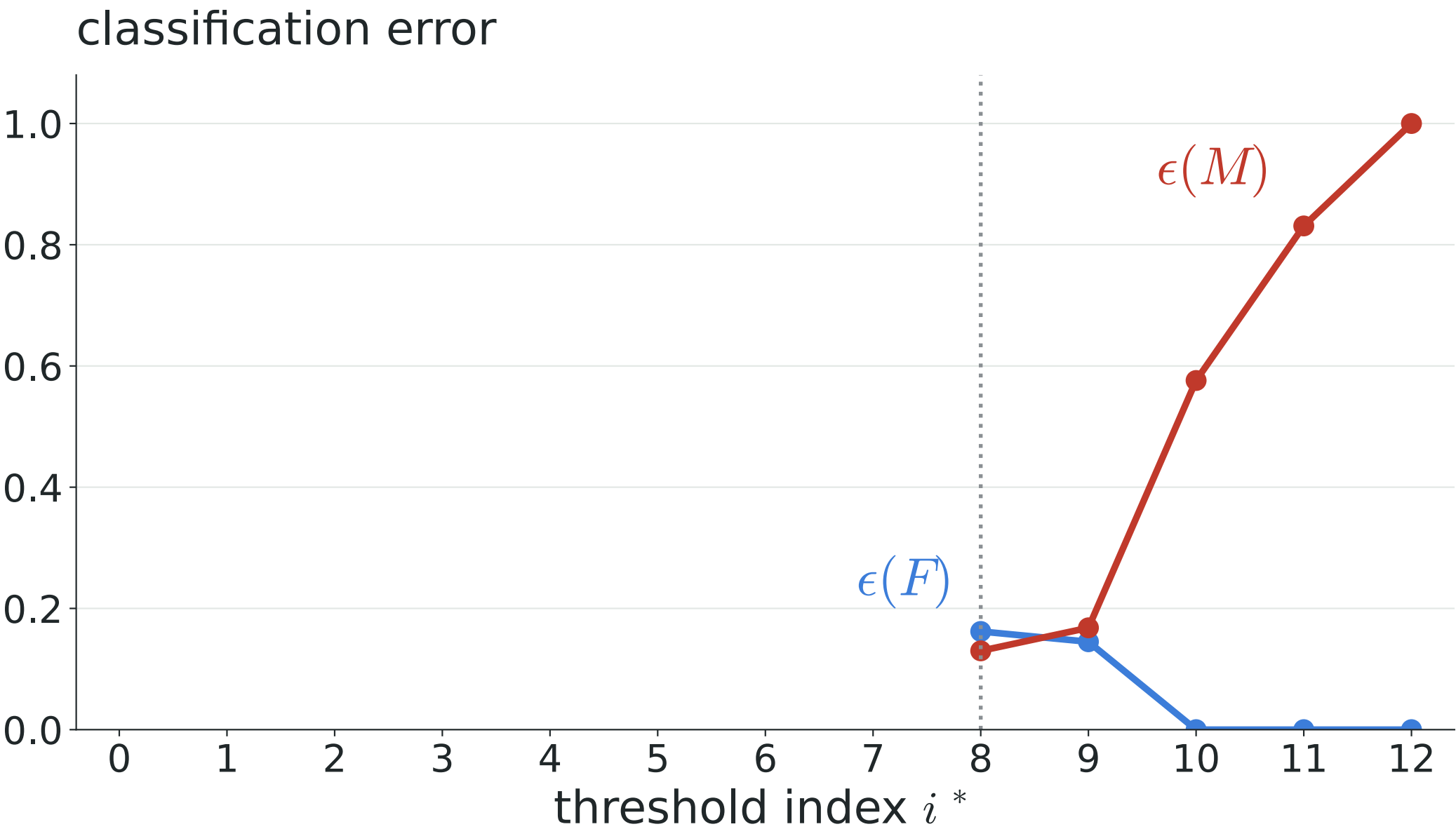
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$\epsilon(F)$									.162	.145	0	0	0
$\epsilon(M)$									.130	.168	.576	.831	1
max									.162	.168	.576	.831	1



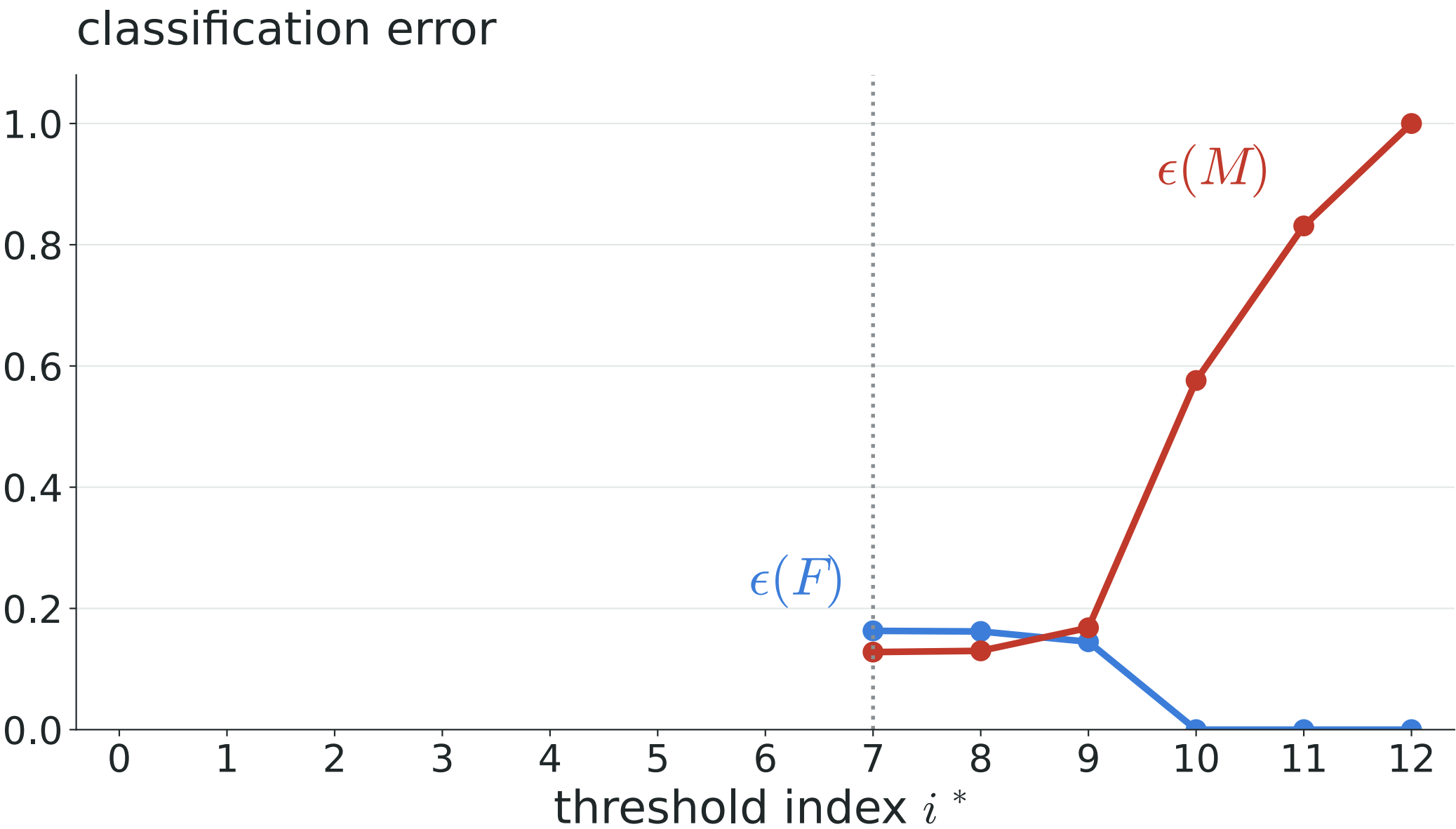
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M if $r \geq$	.034	.056	.065	.117	.237	.647	1.75	2.00	2.24	2.81	∞	∞	never
$\epsilon(F)$								.163	.162	.145	0	0	0
$\epsilon(M)$								.128	.130	.168	.576	.831	1
max								.163	.162	.168	.576	.831	1



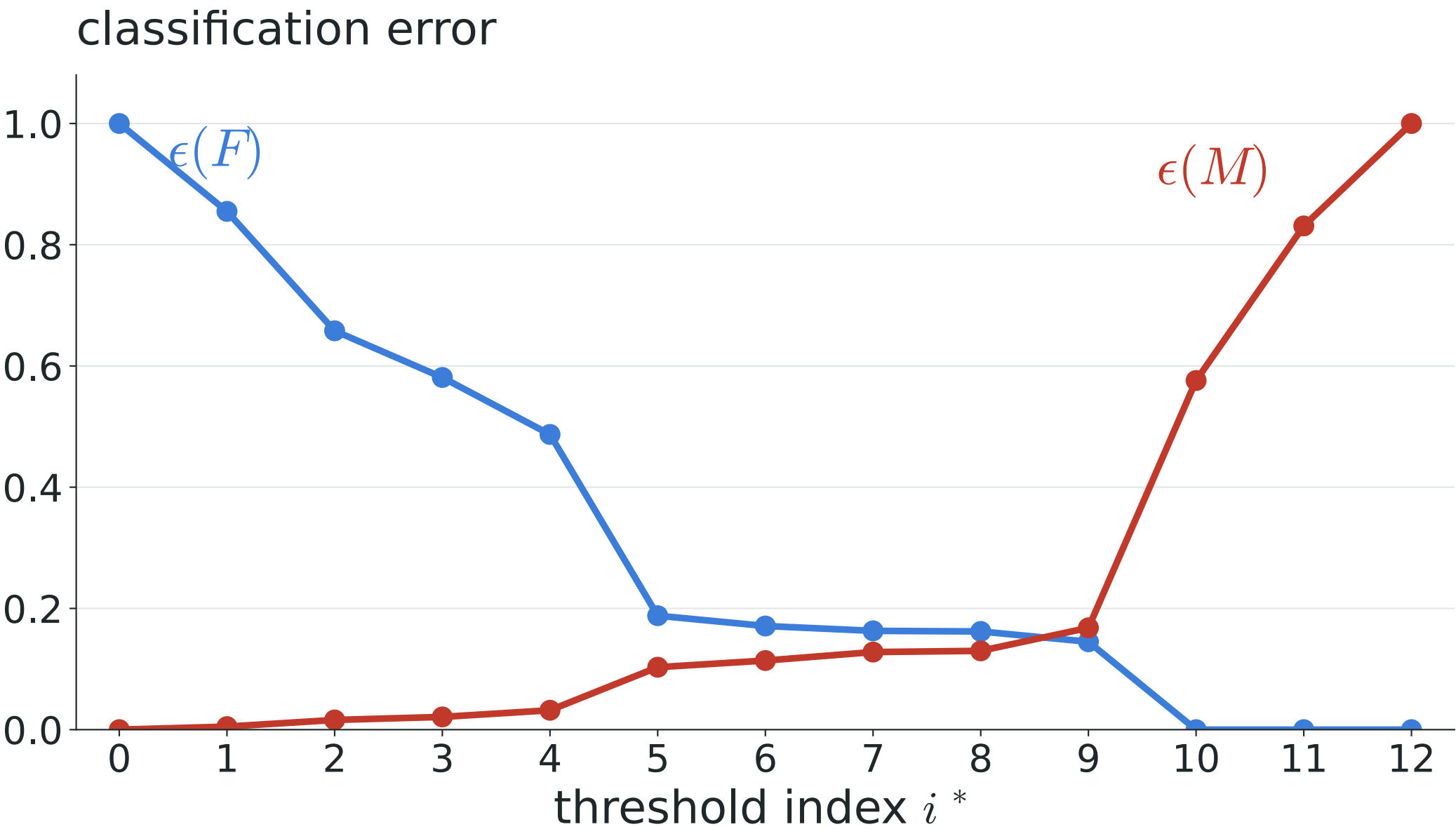
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$\epsilon(F)$	1	.855	.658	.581	.487	.188	.171	.163	.162	.145	0	0	0
$\epsilon(M)$	0	.005	.016	.021	.032	.103	.114	.128	.130	.168	.576	.831	1
max	1	.855	.658	.581	.487	.188	.171	.163	.162	.168	.576	.831	1



Sorted by r , decide M for $i \geq i^*$:

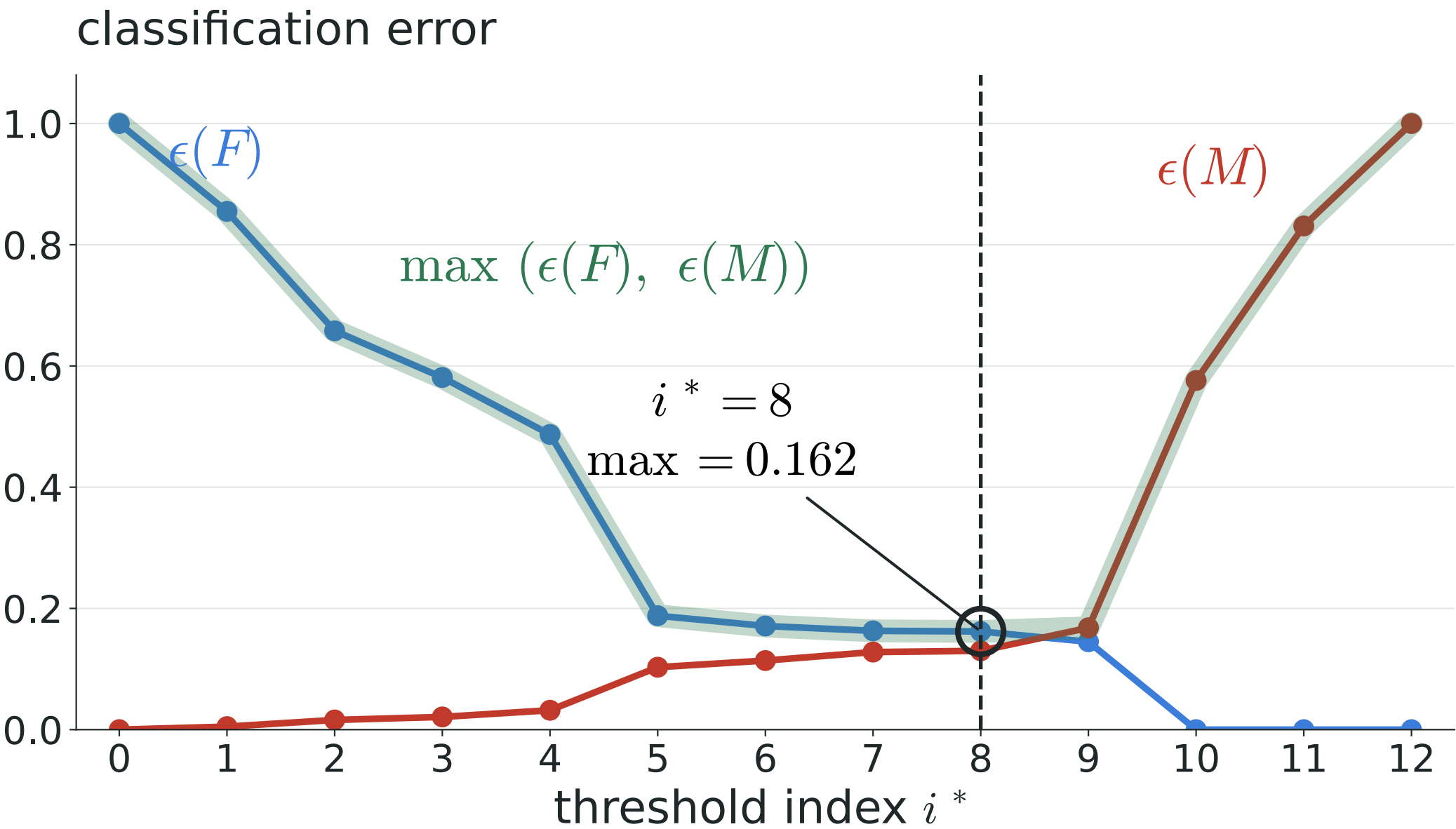
$$\epsilon(F) = \sum_{i \geq i^*} p(x_i | F)$$

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$i^* = 12$: nothing is decided M , i.e. we always decide F .

Derivation: errors for every threshold

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Sorted by r , decide M for $i \geq i^*$:

$$\epsilon(F) = \sum_{i \geq i^*} p(x_i | F)$$
$$\epsilon(M) = \sum_{i < i^*} p(x_i | M)$$

$i^* = 12$: nothing is decided M , i.e. we always decide F .

Smallest maximum: $i^* = 8$

Derivation: the discrete minimax strategy

Strategy $q^*(x)$ for $i^* = 8$

	u-light	light	avg	heavy
short	F	F	F	F
normal	F	F	M	M
tall	F	F	M	M

$$q^*(x) = \begin{cases} M & \text{if } r(x) \geq 2.235 \\ F & \text{otherwise} \end{cases}$$

$$\epsilon(F) = 0.162, \quad \epsilon(M) = 0.130, \quad \max = 0.162$$

! For discrete x the errors usually **cannot be made equal**: take the smallest maximum.

Assignment: ratios 0, ∞ , nan and ties are common. Thresholding the **index** handles them (nan: prefer the first class).

Continuous minimax

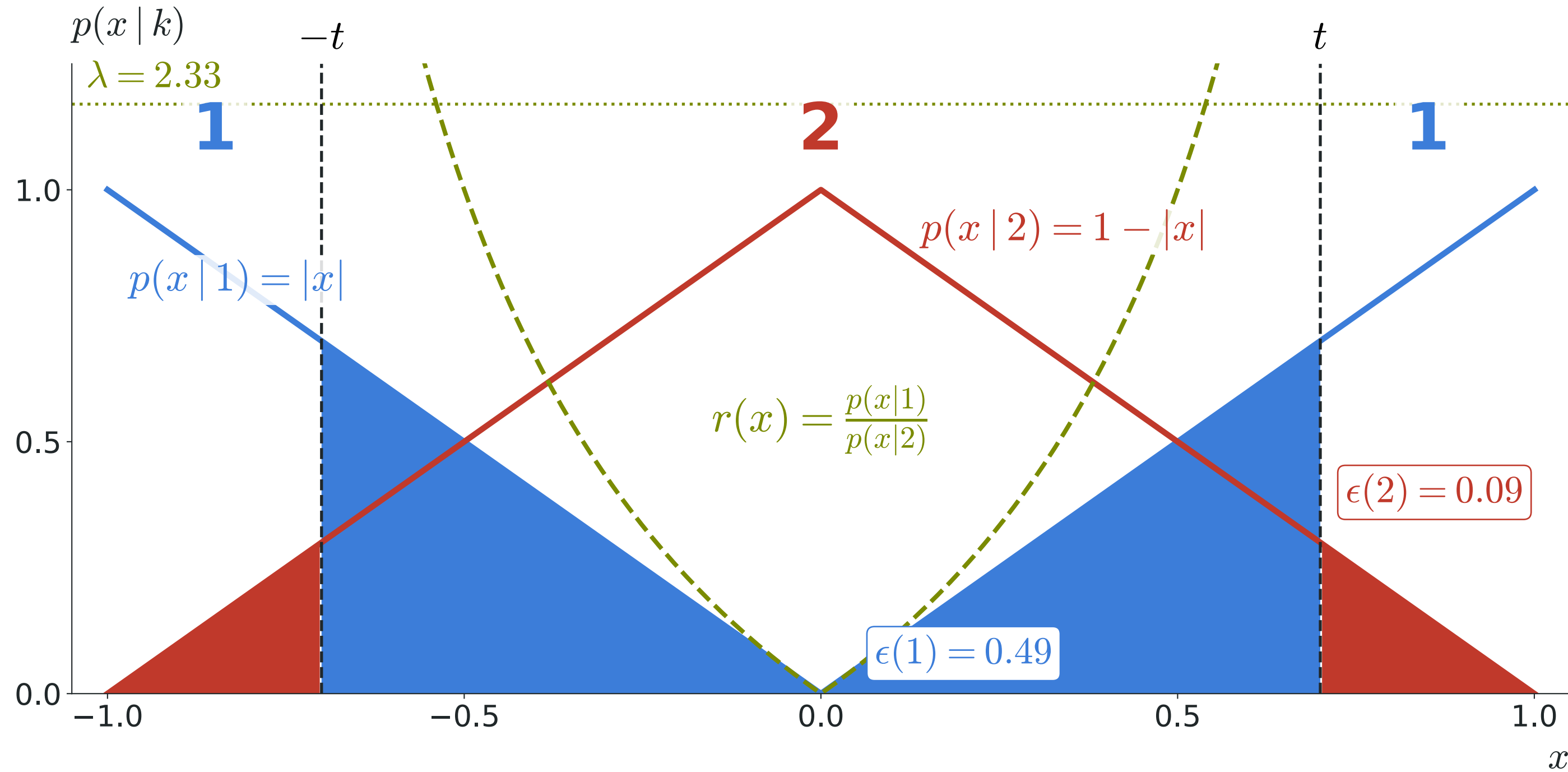
Minimax for continuous x

$$q^* = \arg \min_{q: X \rightarrow K} \max_{k \in K} \epsilon(k), \quad \epsilon(k) = \int_{x: q(x) \neq k} p(x | k) dx$$

? We move the threshold λ . What happens to $\epsilon(1)$, $\epsilon(2)$ and their maximum?

! For continuous x the errors change continuously, one grows and the other shrinks. The optimum is where $\epsilon(1) = \epsilon(2)$.

Example I: two triangles



Step 2: a threshold λ gives the regions **1** | **2** | **1** and the errors. Which λ is optimal?

Example 1: try it

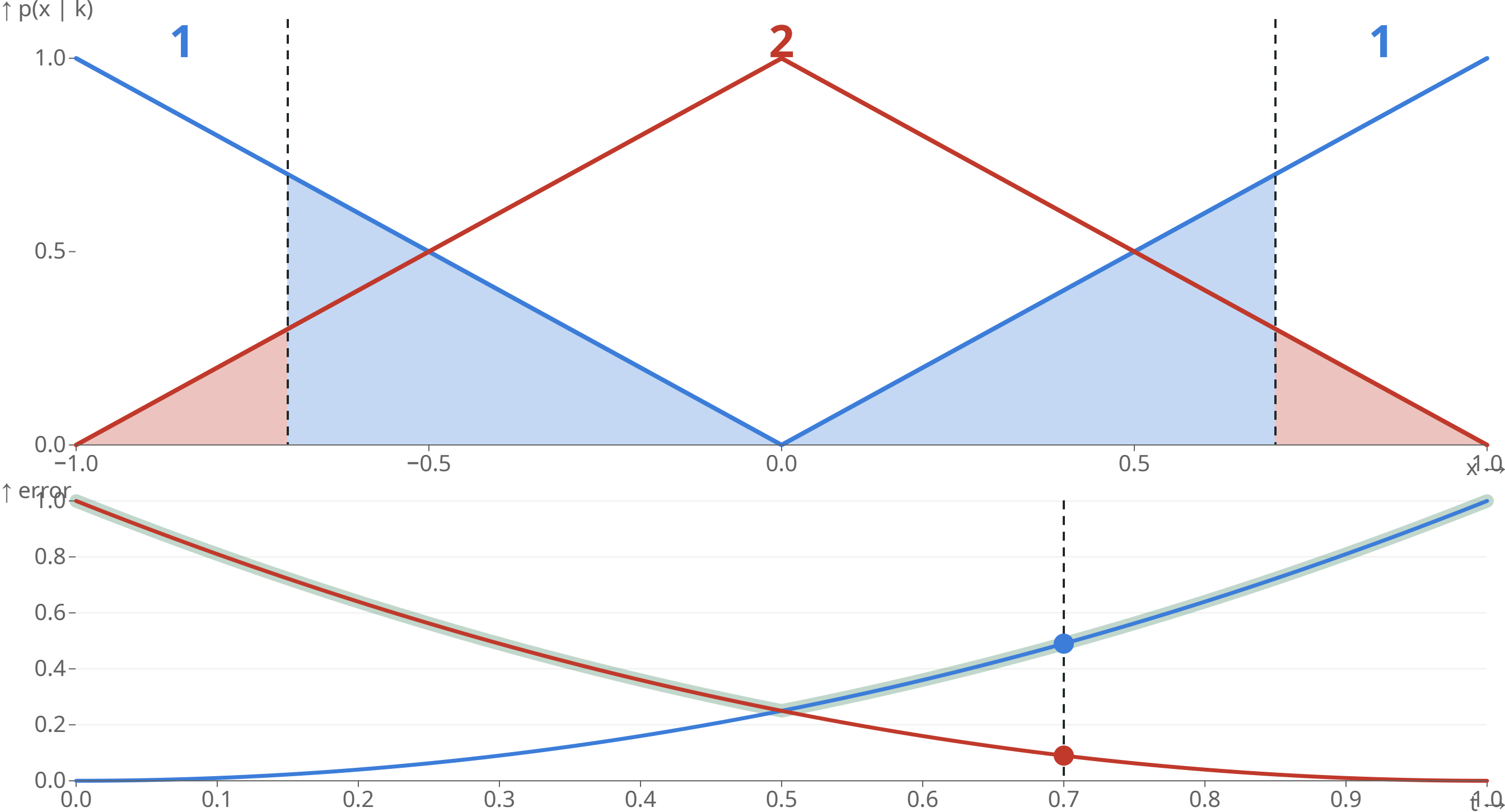
t

$\lambda = 2.33$

$\varepsilon(1) = 0.490$

$\varepsilon(2) = 0.090$

max = 0.490



Derivation: example I

By symmetry the regions are **1** | **2** | **1** with the boundaries $\pm t$:

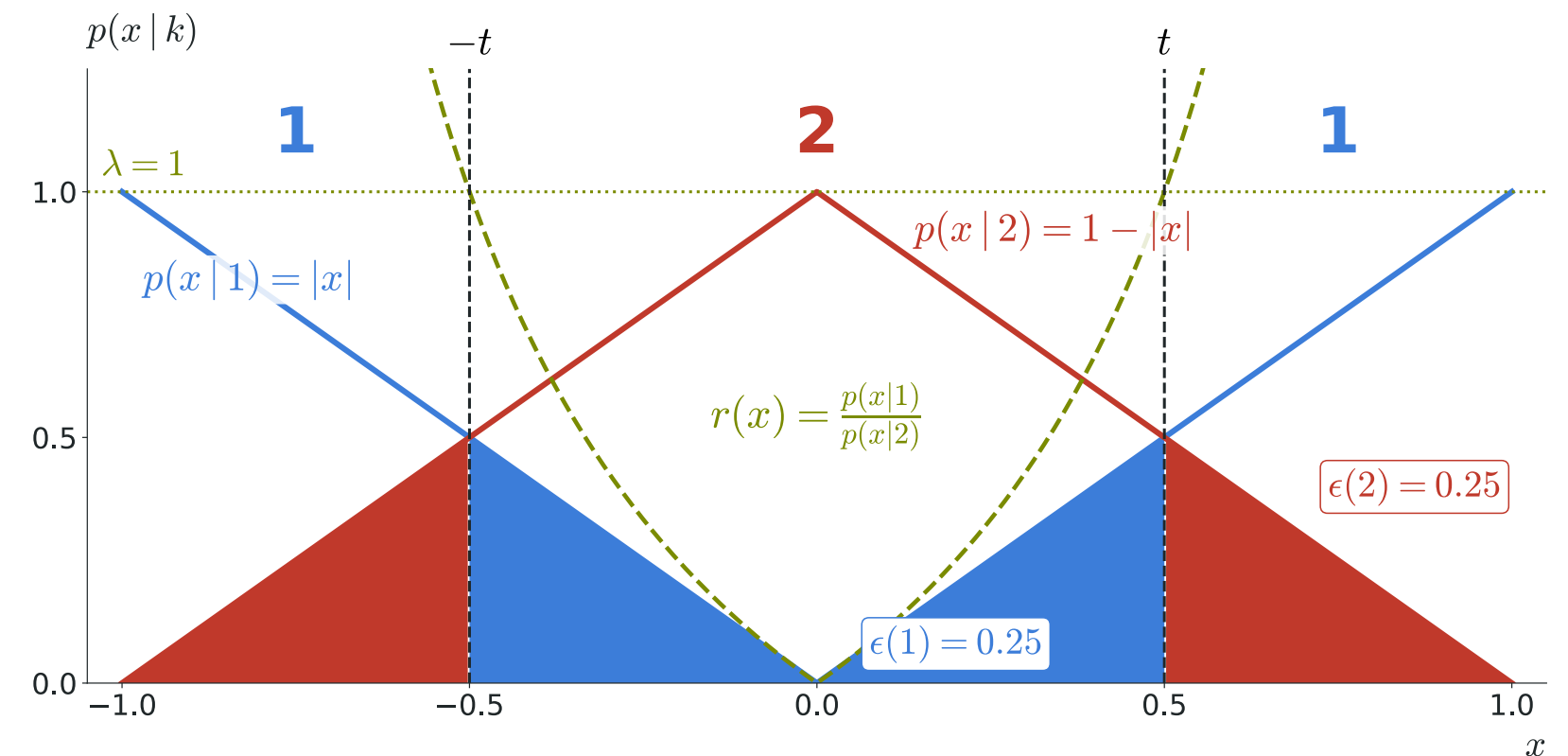
$$\epsilon(\mathbf{1}) = \int_{-t}^t |\mathbf{x}| d\mathbf{x} = 2 \int_0^t \mathbf{x} d\mathbf{x} = t^2$$

$$\epsilon(\mathbf{2}) = 2 \int_t^1 (1 - \mathbf{x}) d\mathbf{x} = (1 - t)^2$$

$$\epsilon(\mathbf{1}) = \epsilon(\mathbf{2}) : \quad t^2 = (1 - t)^2 \quad \text{gives} \quad t = \frac{1}{2}$$

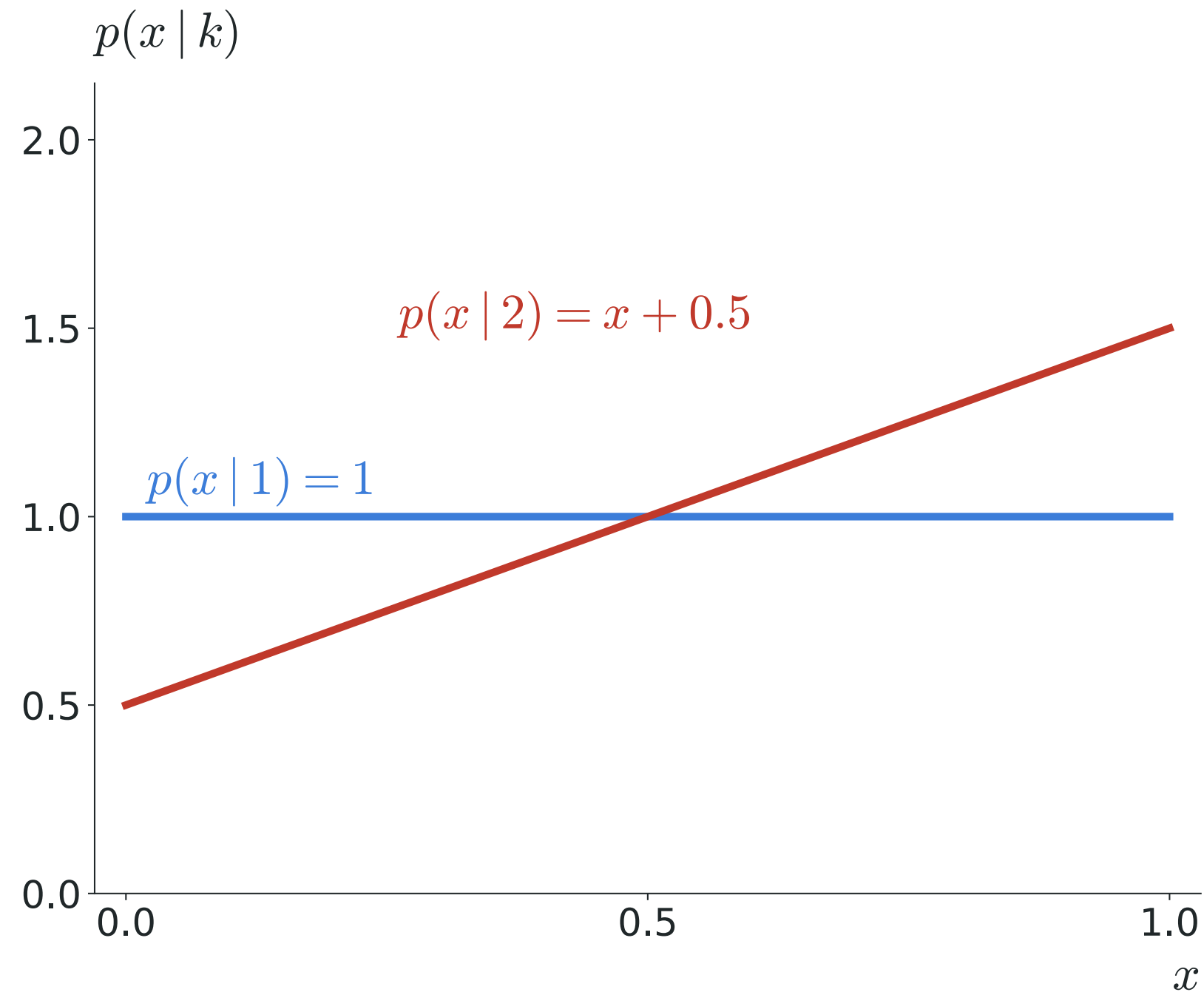
$$\lambda^* = \mathbf{r}\left(\frac{1}{2}\right) = 1, \quad \epsilon(\mathbf{1}) = \epsilon(\mathbf{2}) = \frac{1}{4}$$

$$\mathbf{q}^*(\mathbf{x}) = \begin{cases} \mathbf{2} & \text{if } |\mathbf{x}| \leq \frac{1}{2} \\ \mathbf{1} & \text{otherwise} \end{cases}$$



Example II: your turn

$p(x \mid 1) = 1$, $p(x \mid 2) = x + 0.5$
on the interval $x \in (0, 1)$



- ? 1. What does $r(x)$ look like?
- ? 2. Where do we decide **1** and where **2**?
- ? 3. Which boundary t gives $\epsilon(\mathbf{1}) = \epsilon(\mathbf{2})$?

Derivation: example II

$r(x) = \frac{1}{x+0.5}$ is decreasing: we decide **1** for $x < t$.

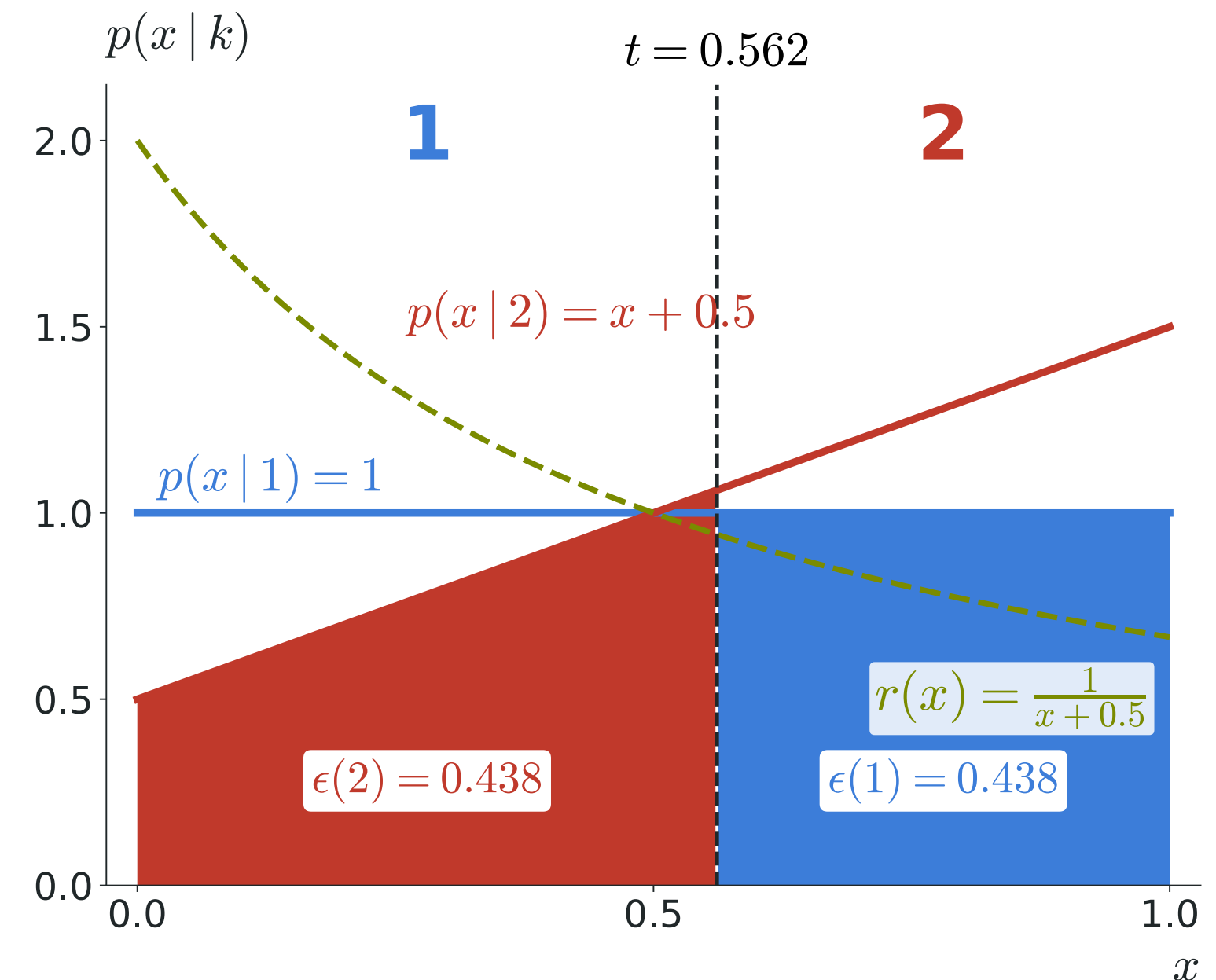
$$\epsilon(\mathbf{1}) = \int_t^1 1 \, dx = 1 - t$$

$$\epsilon(\mathbf{2}) = \int_0^t (x + 0.5) \, dx = \frac{1}{2}t^2 + \frac{1}{2}t$$

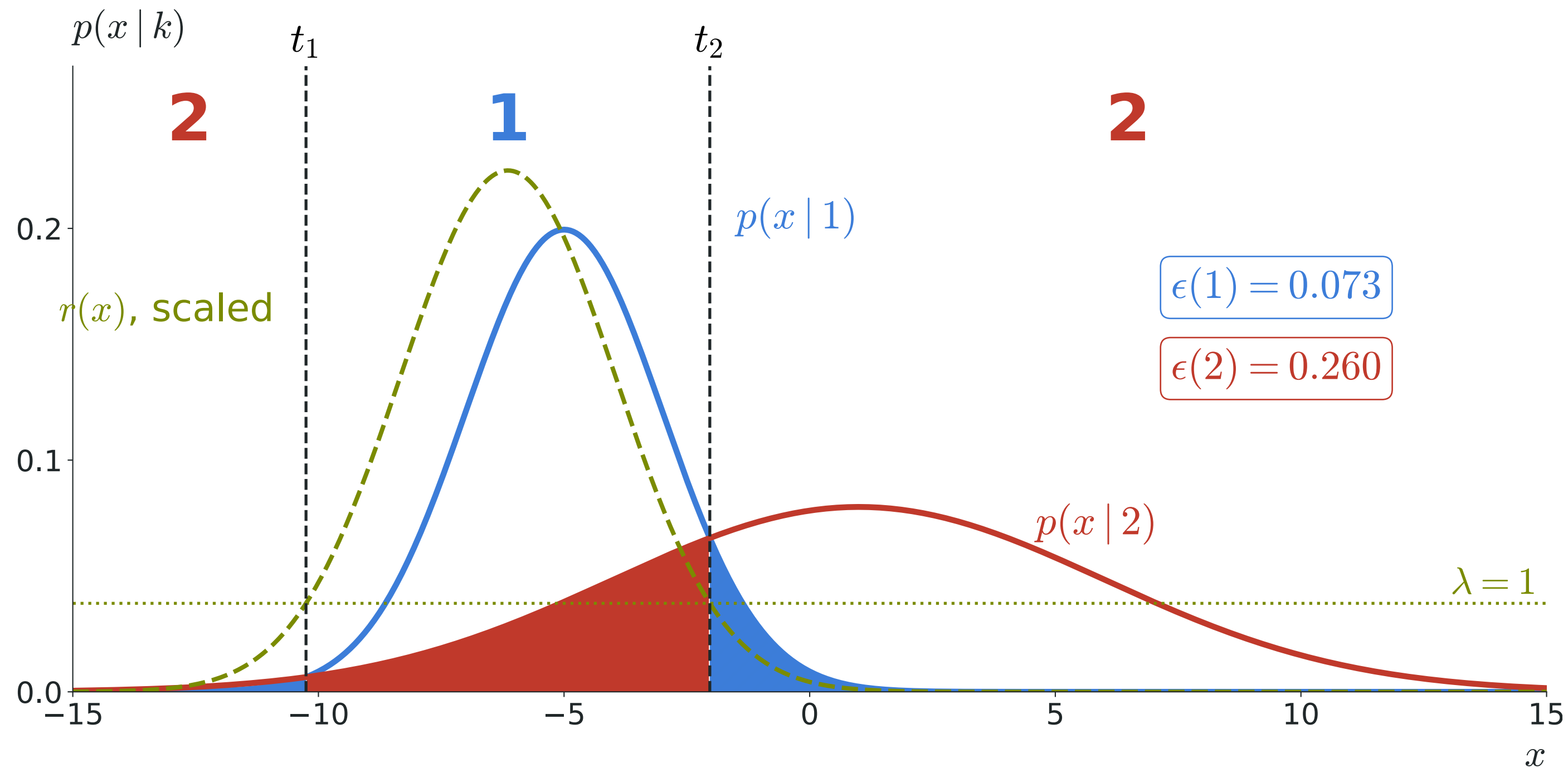
$$t^2 + 3t - 2 = 0, \quad t = \frac{1}{2}(\sqrt{17} - 3) \approx 0.562$$

$$\epsilon(\mathbf{1}) = \epsilon(\mathbf{2}) \approx \mathbf{0.438}, \quad \lambda^* = r(t) \approx 0.942$$

$$q^*(x) = \begin{cases} \mathbf{1} & \text{if } x < 0.562 \\ \mathbf{2} & \text{otherwise} \end{cases}$$



Example III: two normal distributions



Regions **2** | **1** | **2** and the errors for $\lambda = 1$. How do we get $\epsilon(\mathbf{1}) = \epsilon(\mathbf{2})$?

Derivation: from λ to the boundaries

We decide 1 where $\frac{p(x | 1)}{p(x | 2)} \geq \lambda$:

$$\frac{\sigma_2}{\sigma_1} \exp\left(-\frac{(x - \mu_1)^2}{2\sigma_1^2} + \frac{(x - \mu_2)^2}{2\sigma_2^2}\right) \geq \lambda$$

Take the logarithm: the same inequality $ax^2 + bx + c \geq 0$ as in Lab 2

$$-\frac{(x - \mu_1)^2}{2\sigma_1^2} + \frac{(x - \mu_2)^2}{2\sigma_2^2} \geq \ln\left(\frac{\sigma_1}{\sigma_2} \cdot \lambda\right) =: L$$

$$a = \sigma_1^2 - \sigma_2^2$$

$$b = 2(\sigma_2^2 \mu_1 - \sigma_1^2 \mu_2)$$

$$c = \sigma_1^2 \mu_2^2 - \sigma_2^2 \mu_1^2 - 2\sigma_1^2 \sigma_2^2 L$$

! Lab 2 had $\frac{p(2)}{p(1)}$ in place of λ : this is the **Bayesian strategy** for $p(1) = \frac{1}{1 + \lambda}$

Derivation: the errors of the strategy

For $\sigma_1 < \sigma_2$ the regions are $\textcolor{red}{2} \mid \textcolor{blue}{1} \mid \textcolor{red}{2}$ with the boundaries $t_1 < t_2$.

With the distribution functions F_1, F_2 (`norm.cdf`):

$$\epsilon(\textcolor{blue}{1}) = \int_{-\infty}^{t_1} p(\textcolor{teal}{x} \mid \textcolor{blue}{1}) d\textcolor{teal}{x} + \int_{t_2}^{\infty} p(\textcolor{teal}{x} \mid \textcolor{blue}{1}) d\textcolor{teal}{x} = 1 - (F_1(t_2) - F_1(t_1))$$

$$\epsilon(\textcolor{red}{2}) = \int_{t_1}^{t_2} p(\textcolor{teal}{x} \mid \textcolor{red}{2}) d\textcolor{teal}{x} = F_2(t_2) - F_2(t_1)$$

 $\epsilon(\textcolor{blue}{1}) = \epsilon(\textcolor{red}{2})$ has no closed-form solution for normal distributions. We have to search for λ numerically.

Minimax and Bayes

The risk of a fixed strategy

? We fix a strategy q . How does its risk change when the prior changes?

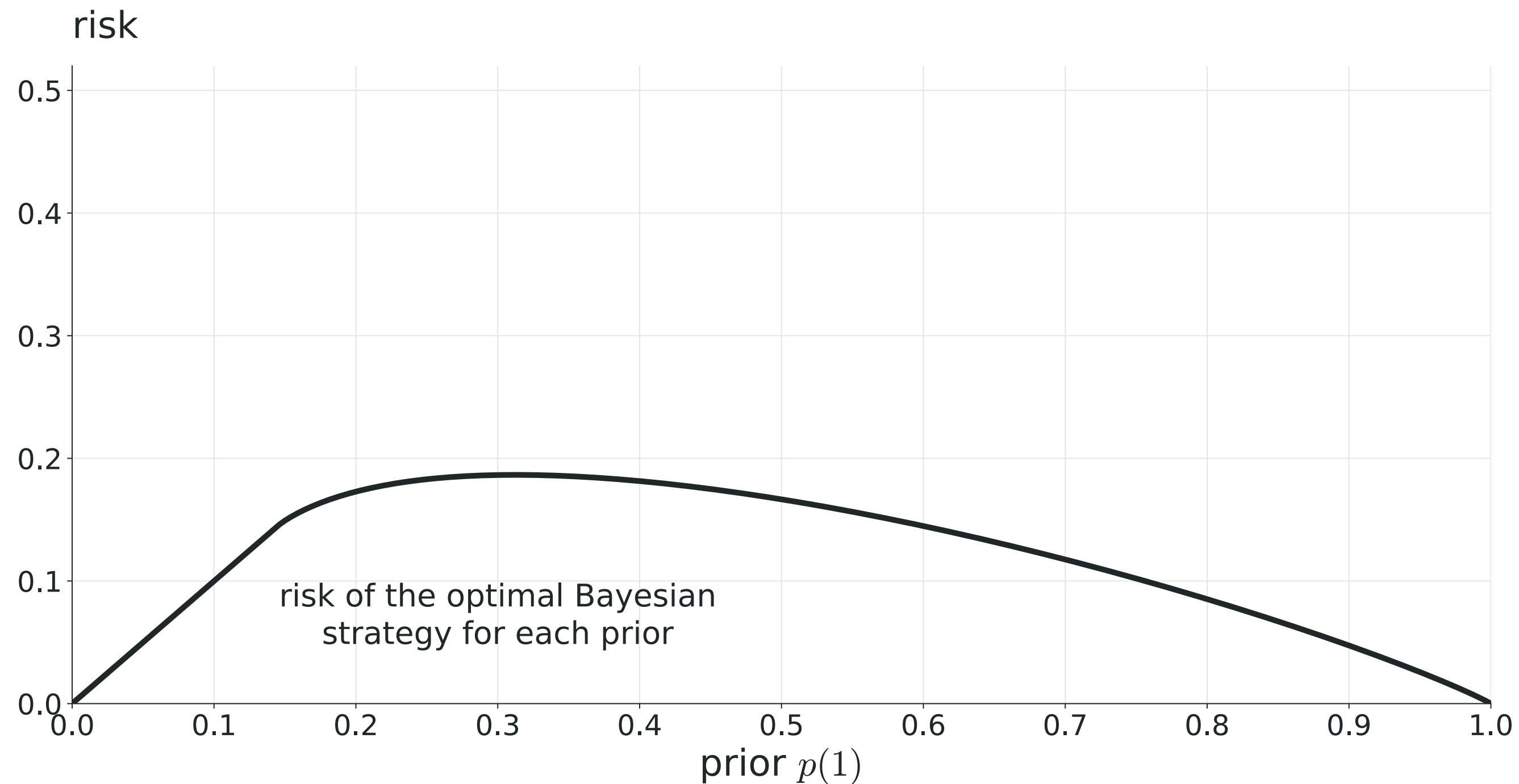
$$R(q) = \sum_k \int_{x: q(x) \neq k} p_{XK}(x, k) dx = \sum_k p(k) \underbrace{\int_{x: q(x) \neq k} p(x | k) dx}_{\epsilon(k)}$$

For two classes, with $p(2) = 1 - p(1)$:

$$R(q) = p(1) \epsilon(1) + (1 - p(1)) \epsilon(2) = p(1) (\epsilon(1) - \epsilon(2)) + \epsilon(2)$$

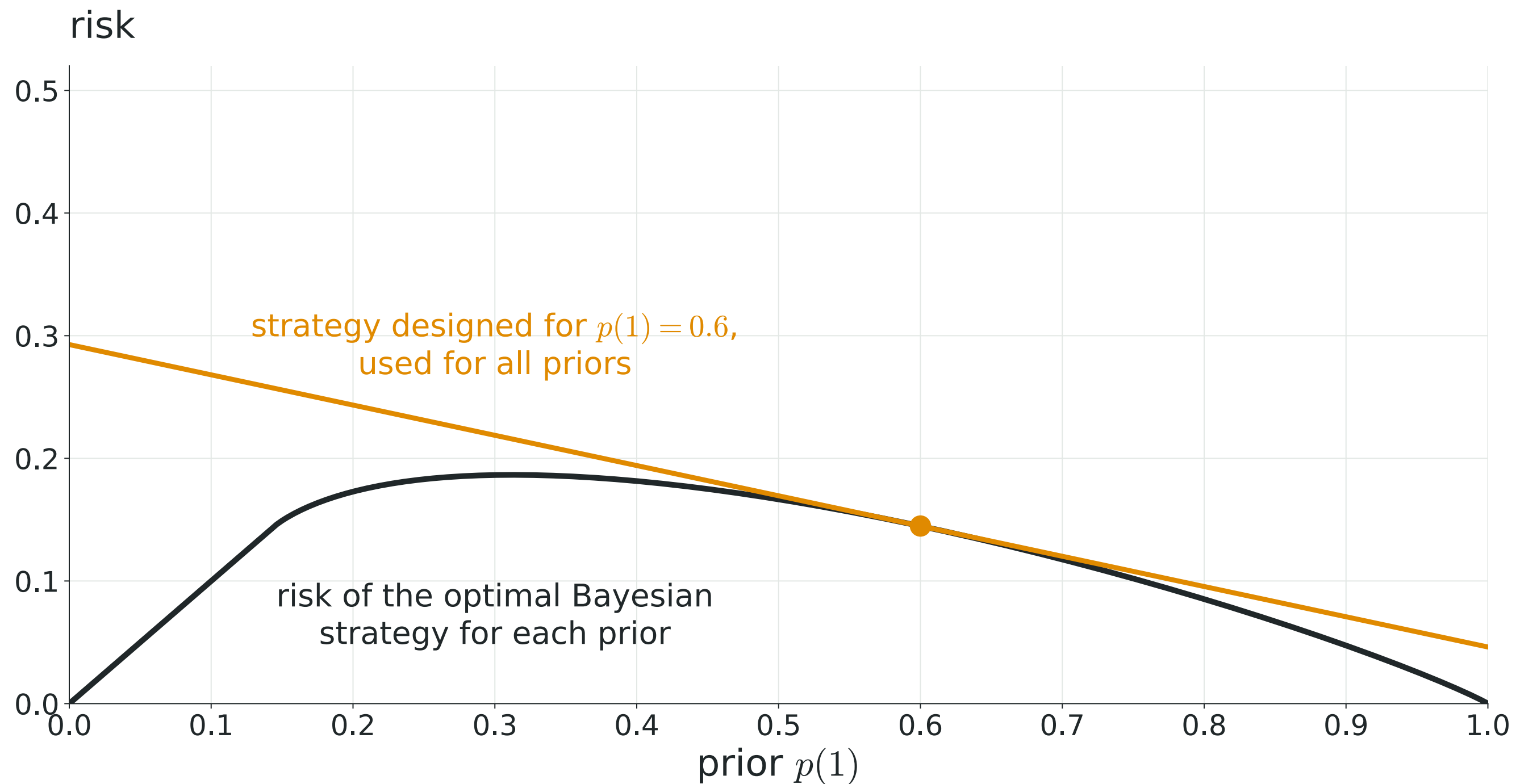
! **Linear** in $p(1)$, from $\epsilon(2)$ at $p(1) = 0$ to $\epsilon(1)$ at $p(1) = 1$. Worst case: $\max(\epsilon(1), \epsilon(2))$

Minimax as the worst-case Bayes



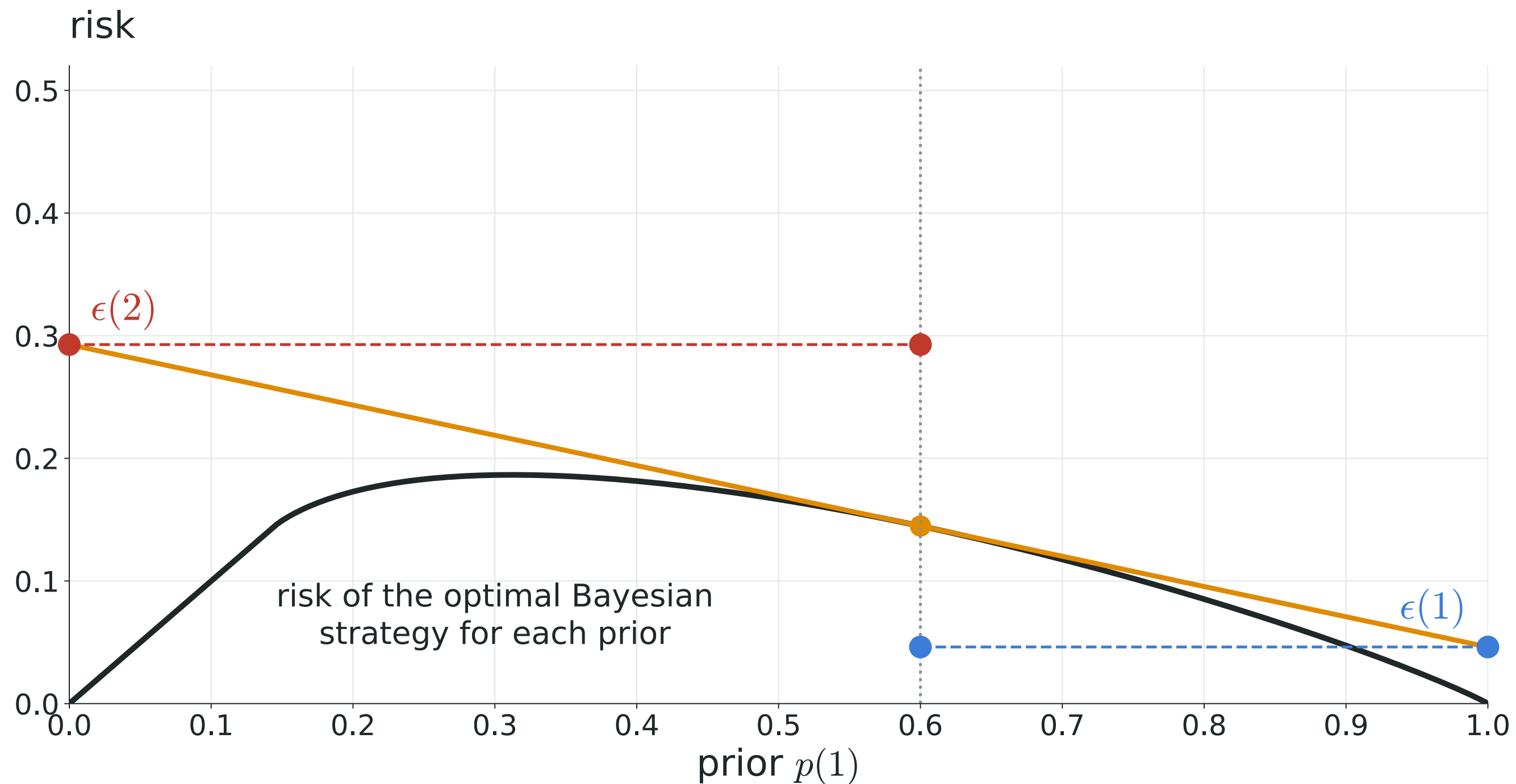
For each prior $p(1)$: the risk of **its own** optimal Bayesian strategy (example III)

Minimax as the worst-case Bayes



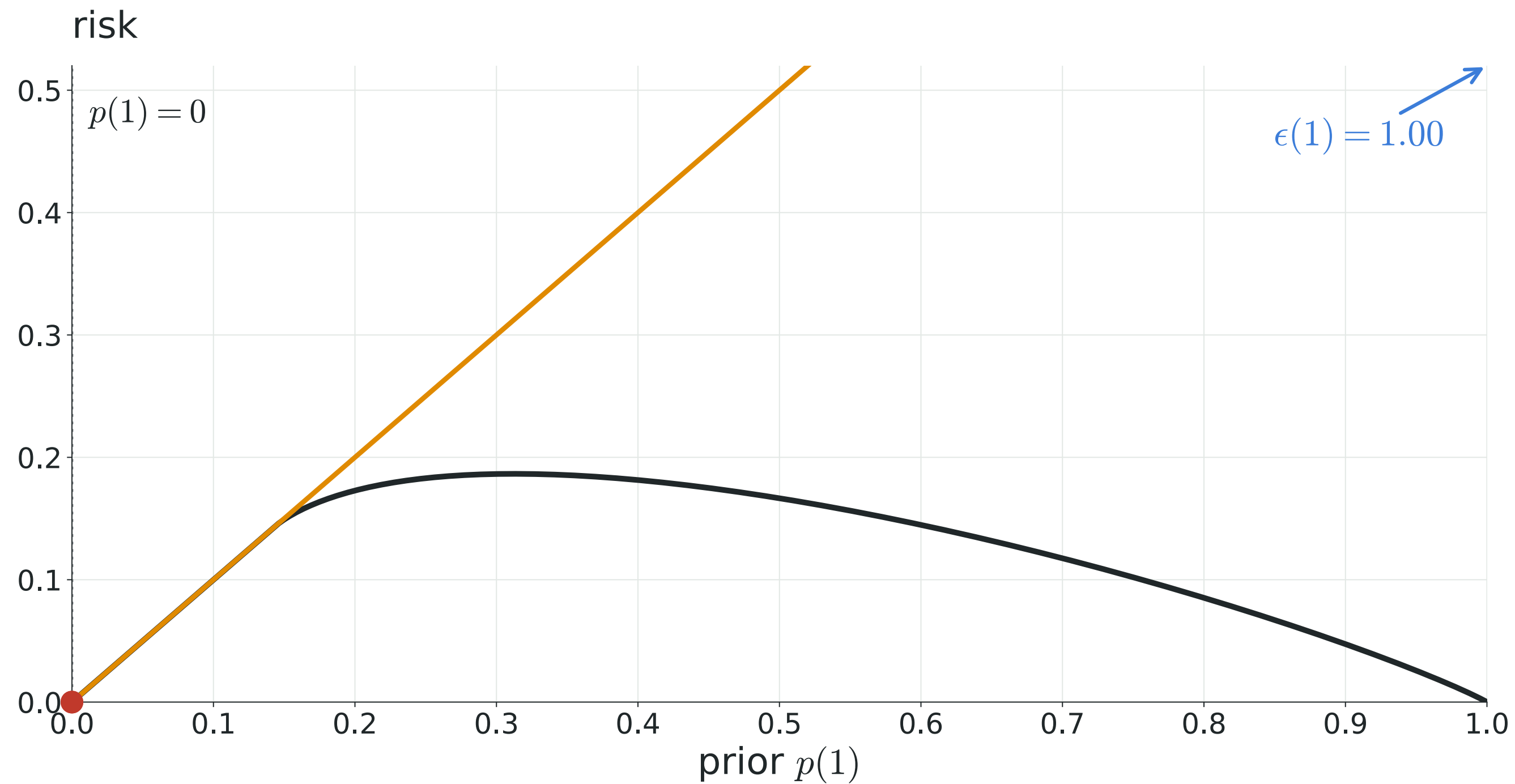
The strategy designed for $p(1) = 0.6$, used everywhere: a line touching the curve

Minimax as the worst-case Bayes



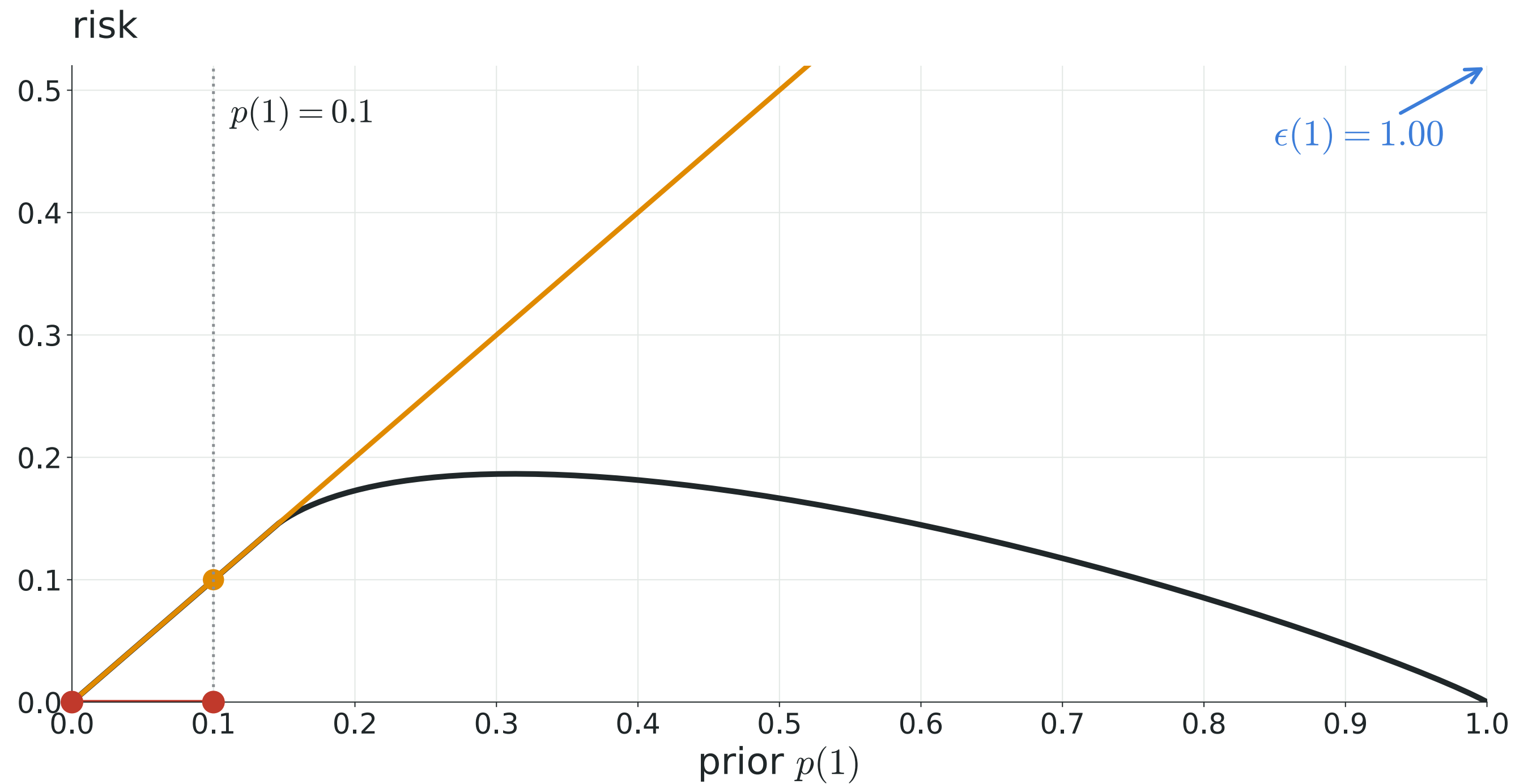
Its ends are $\epsilon(2)$ (left) and $\epsilon(1)$ (right). We mark both above its prior

Minimax as the worst-case Bayes



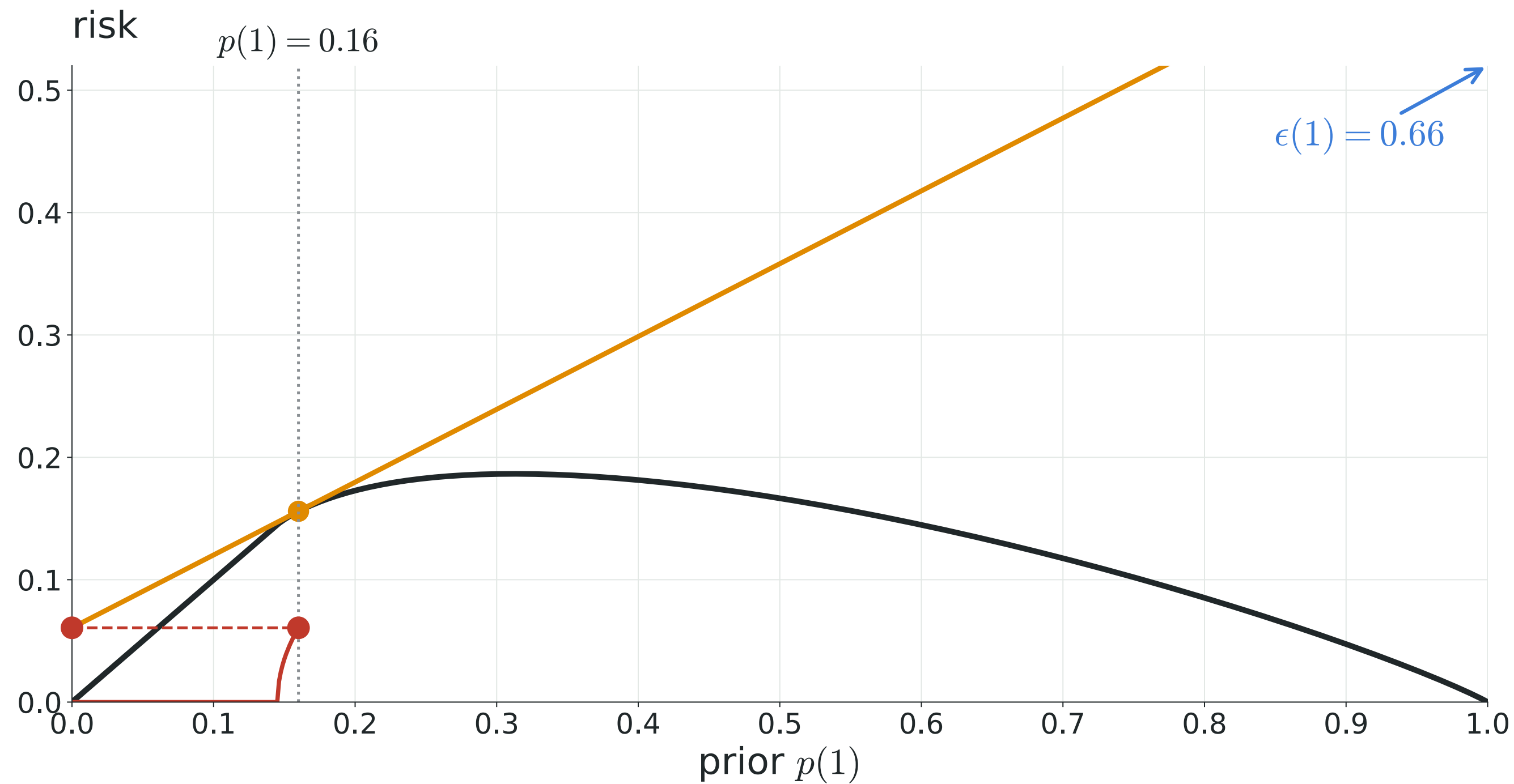
The same for the strategies of all priors from 0 to 1: the points draw $\epsilon(1)$ and $\epsilon(2)$

Minimax as the worst-case Bayes



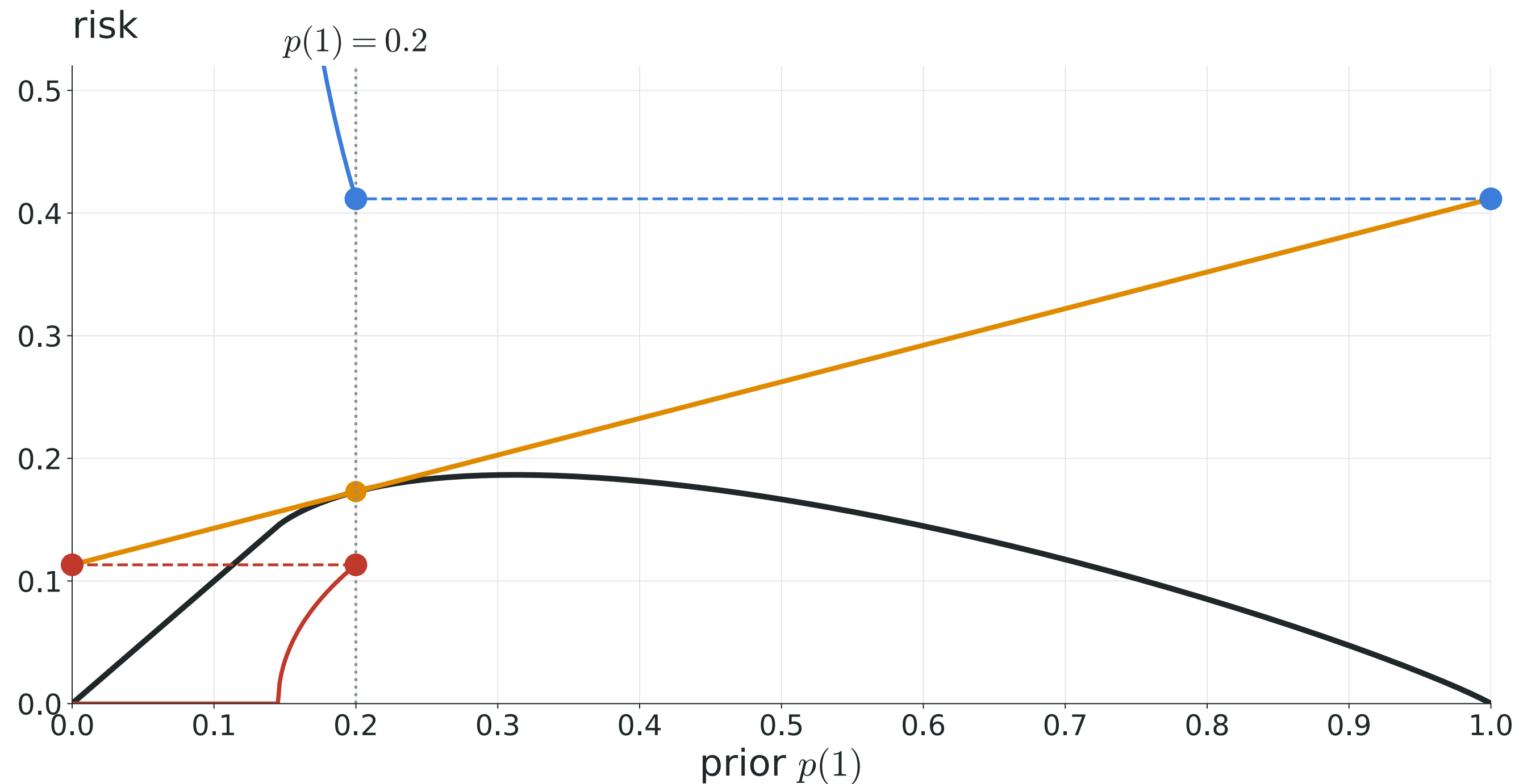
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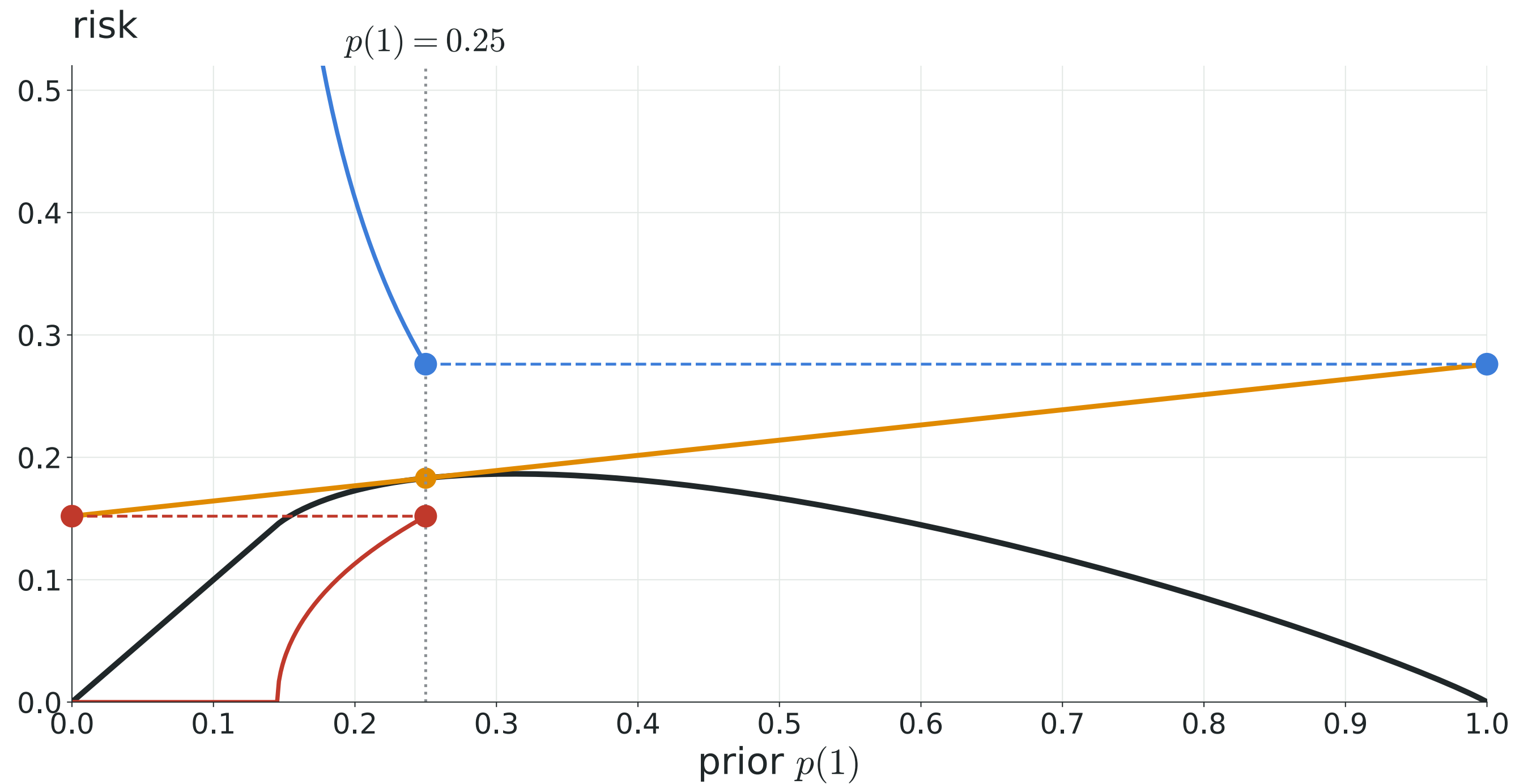
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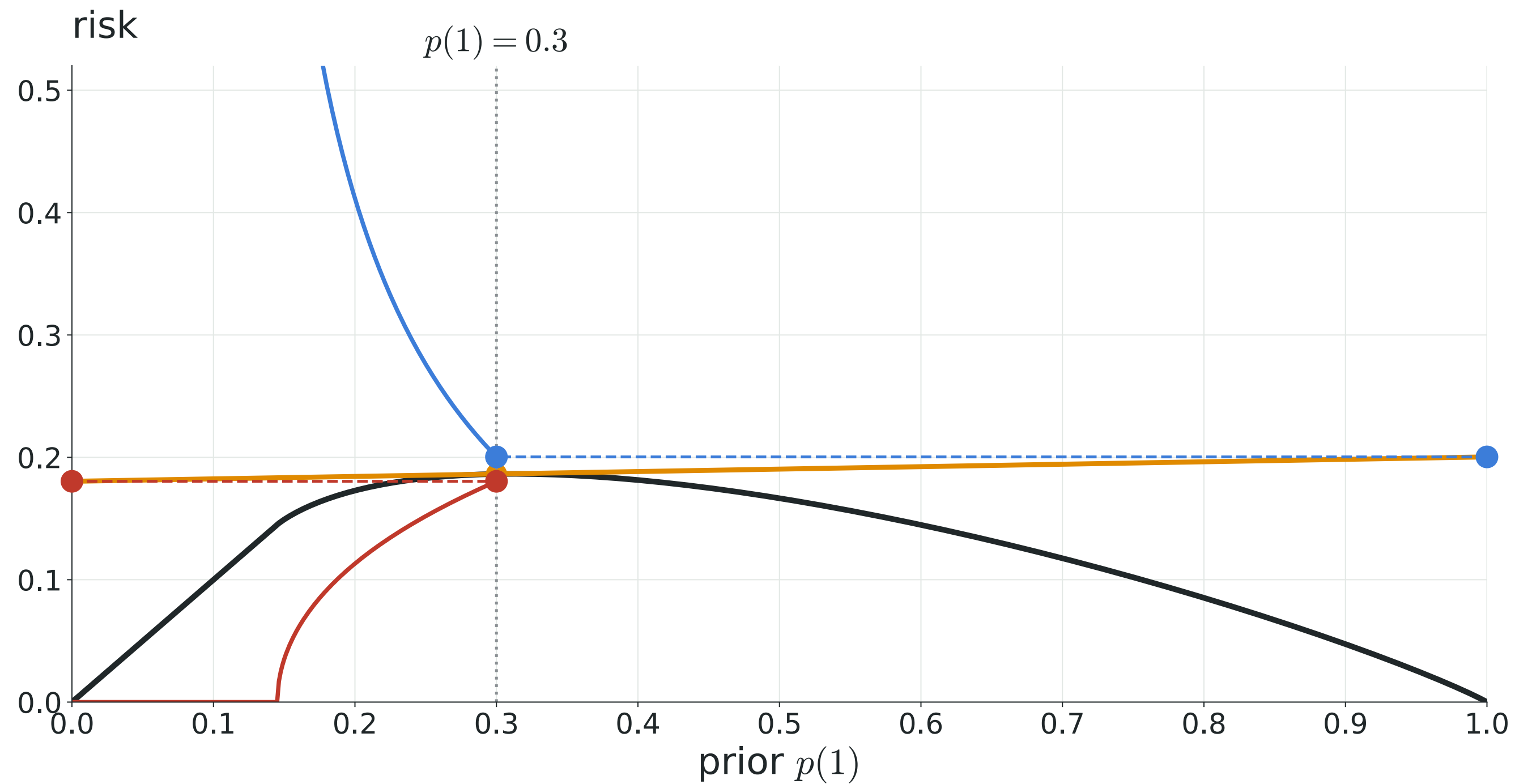
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Minimax as the worst-case Bayes



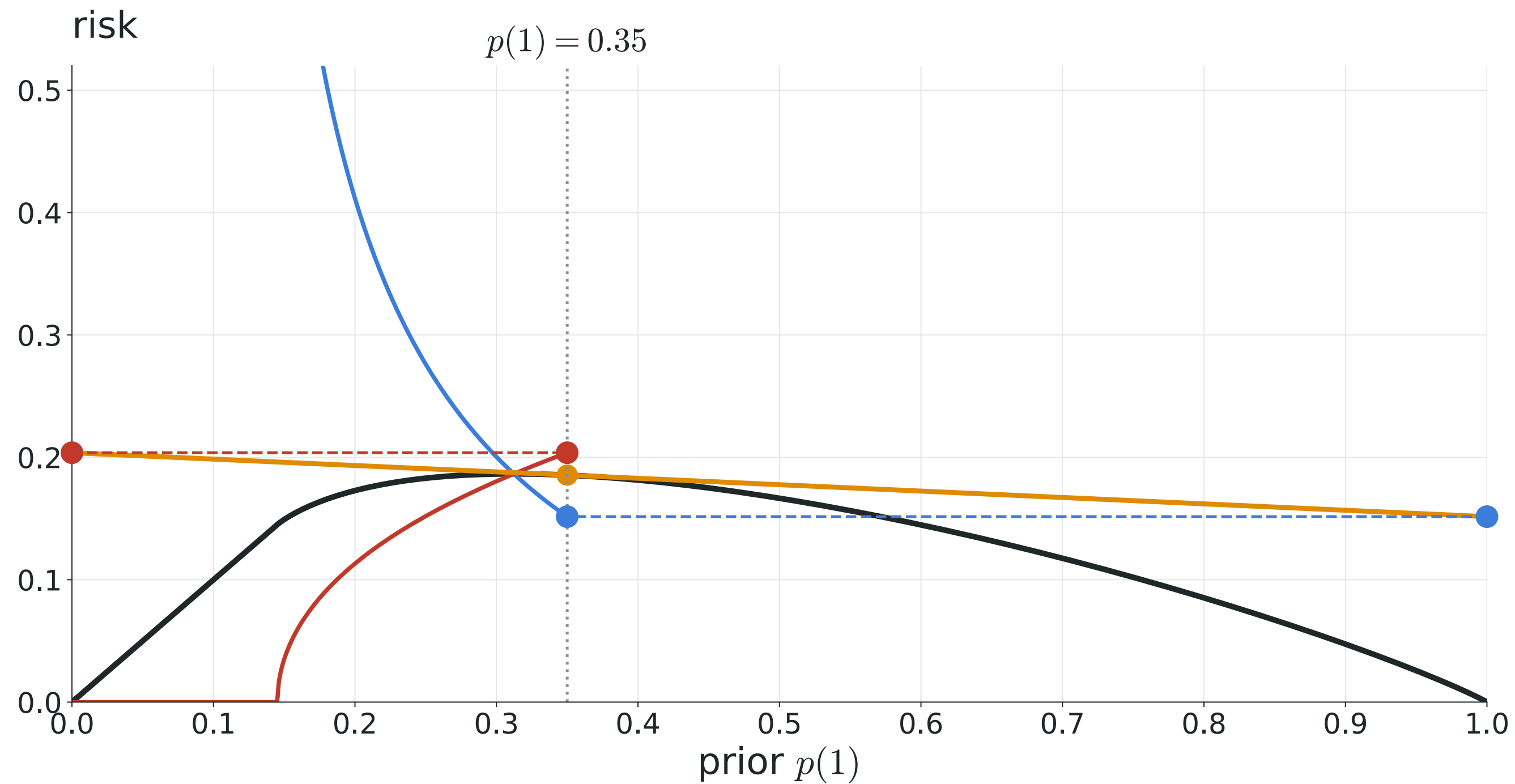
The same for the strategies of all priors from 0 to 1: the points draw $\epsilon(1)$ and $\epsilon(2)$

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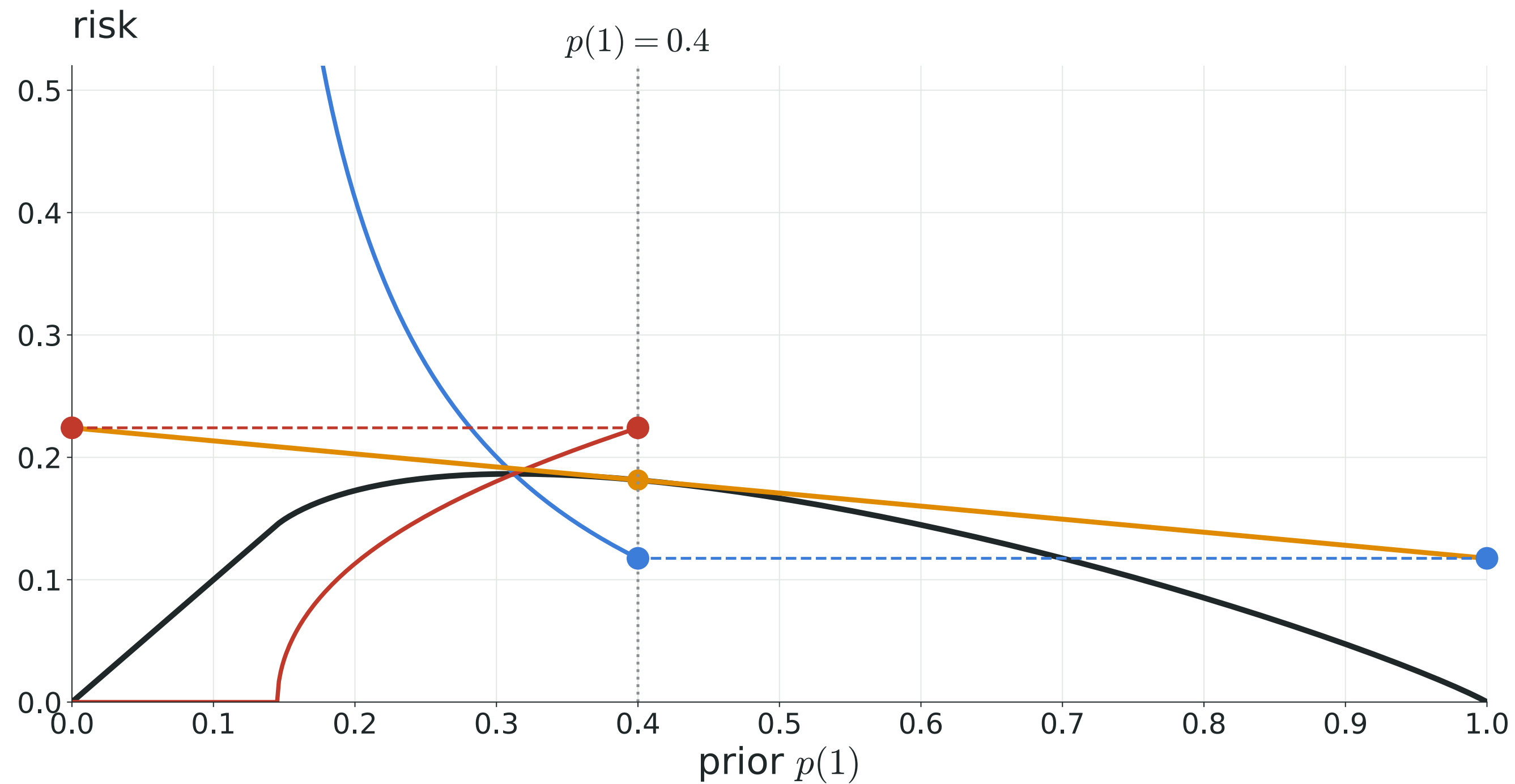
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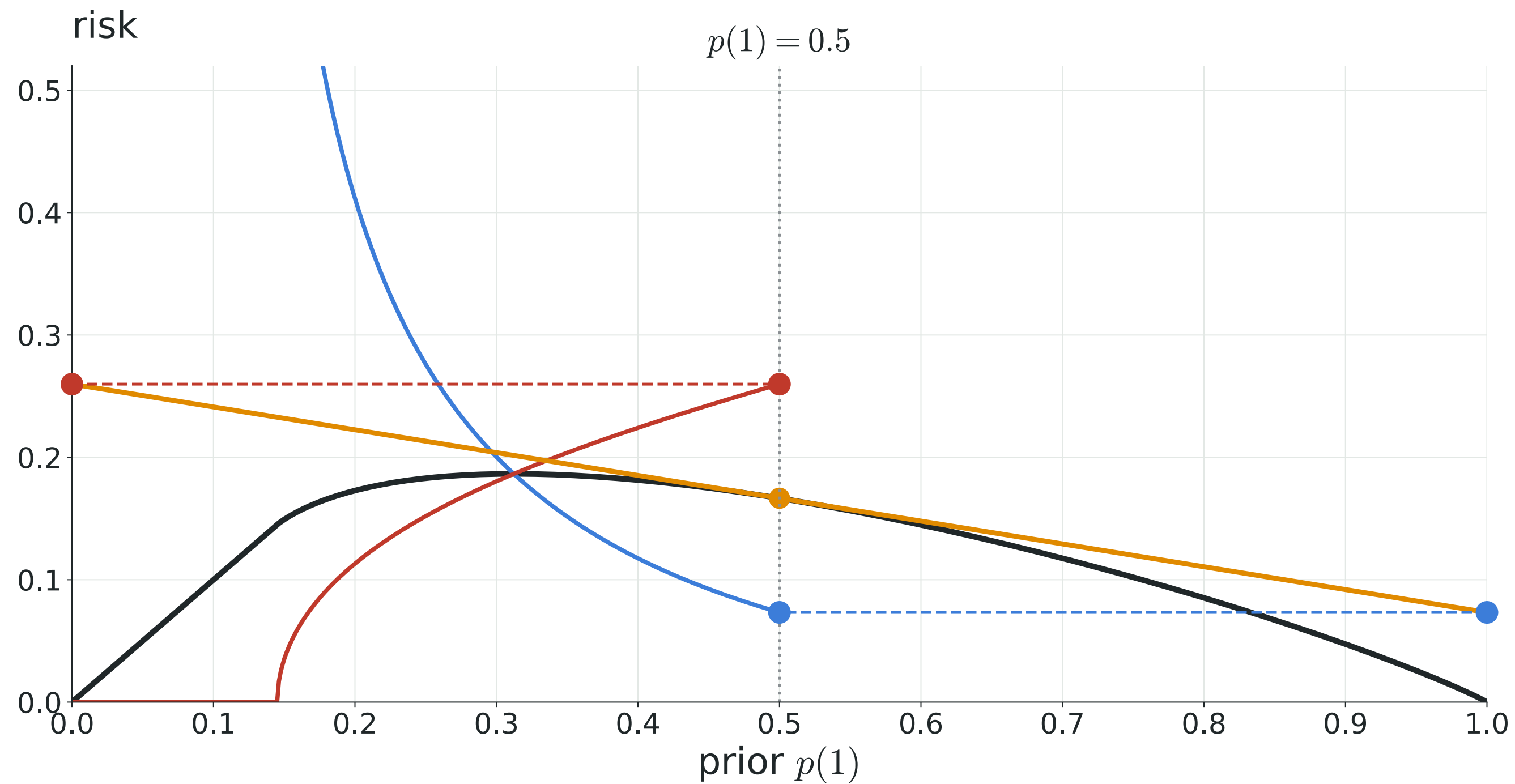
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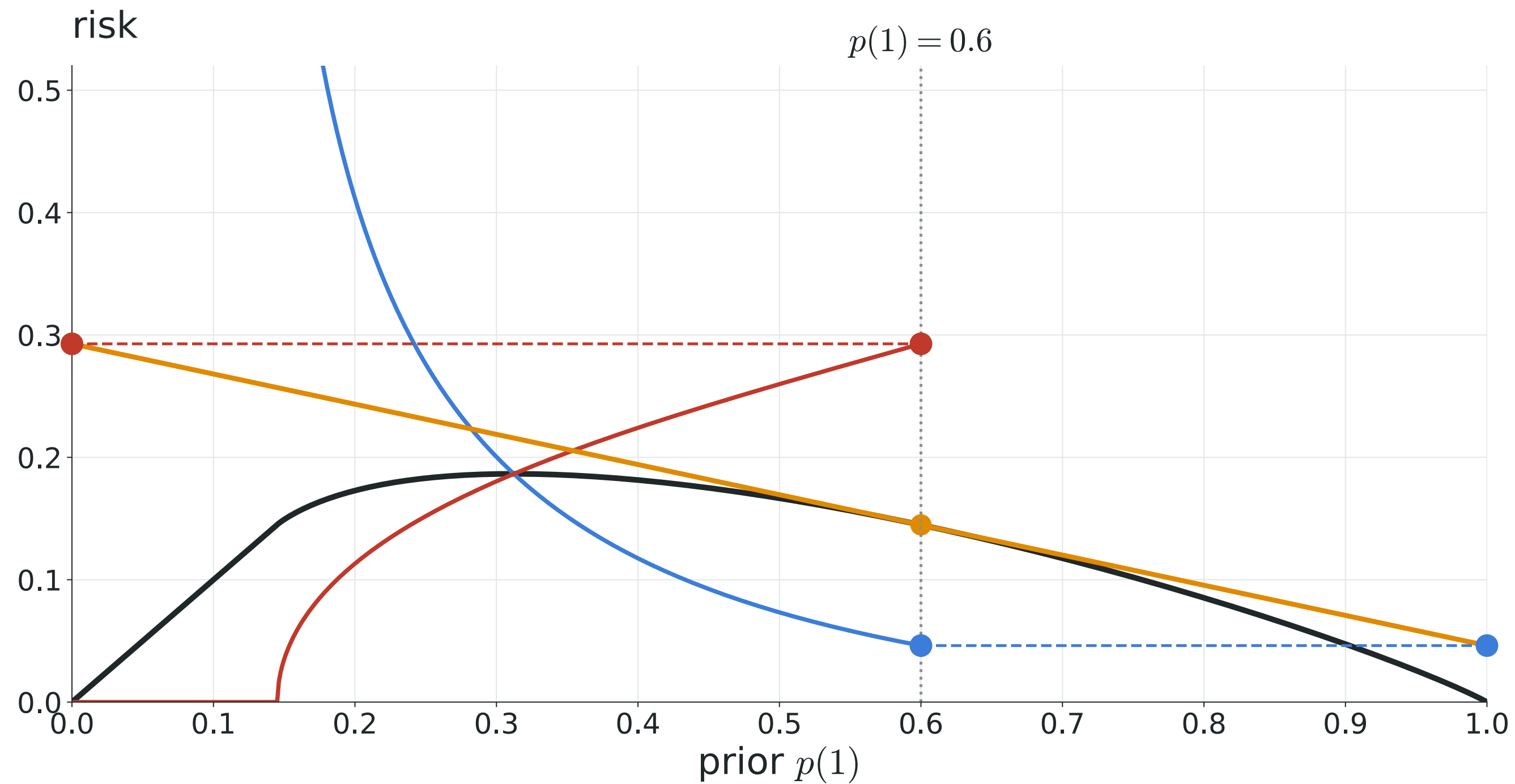
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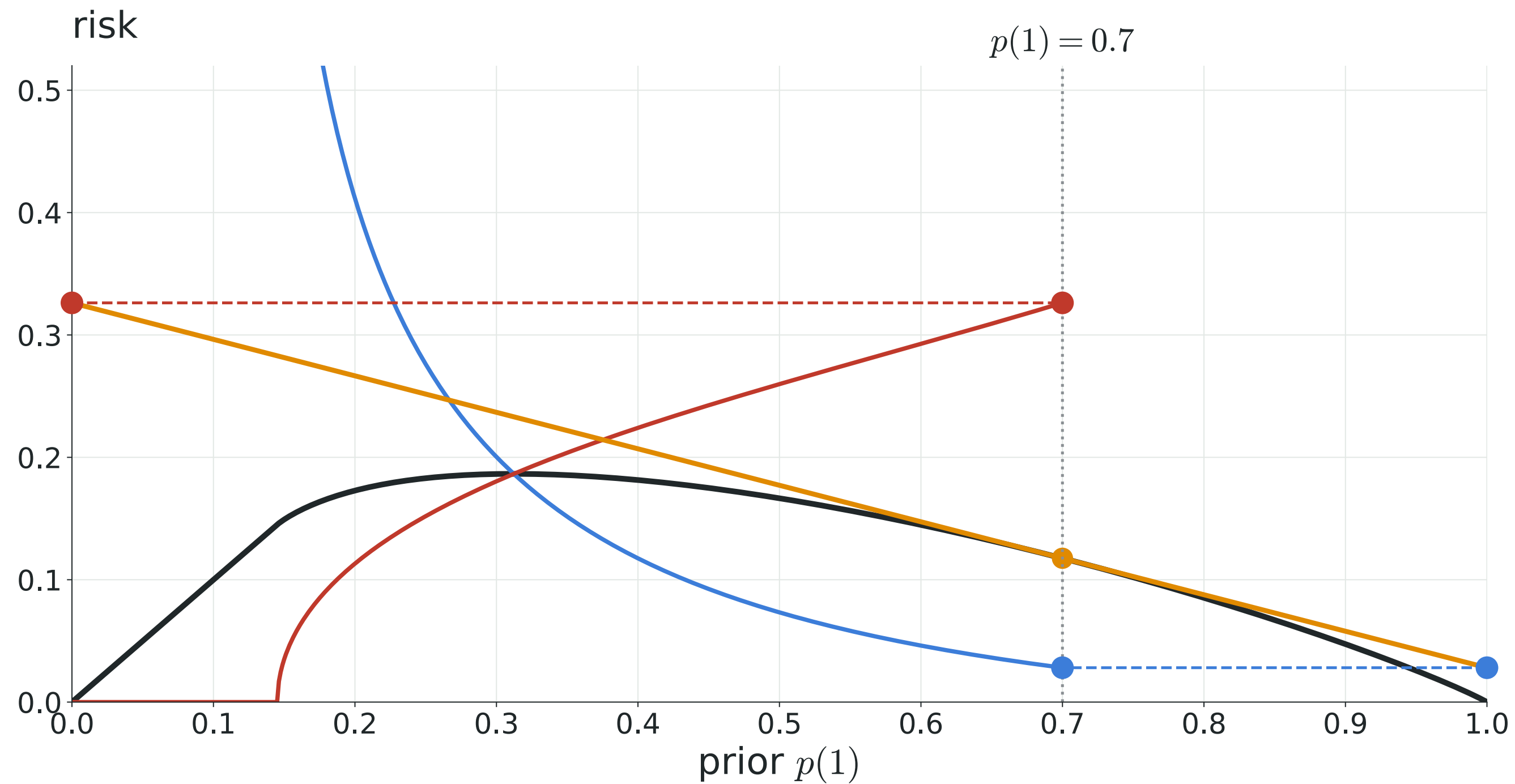
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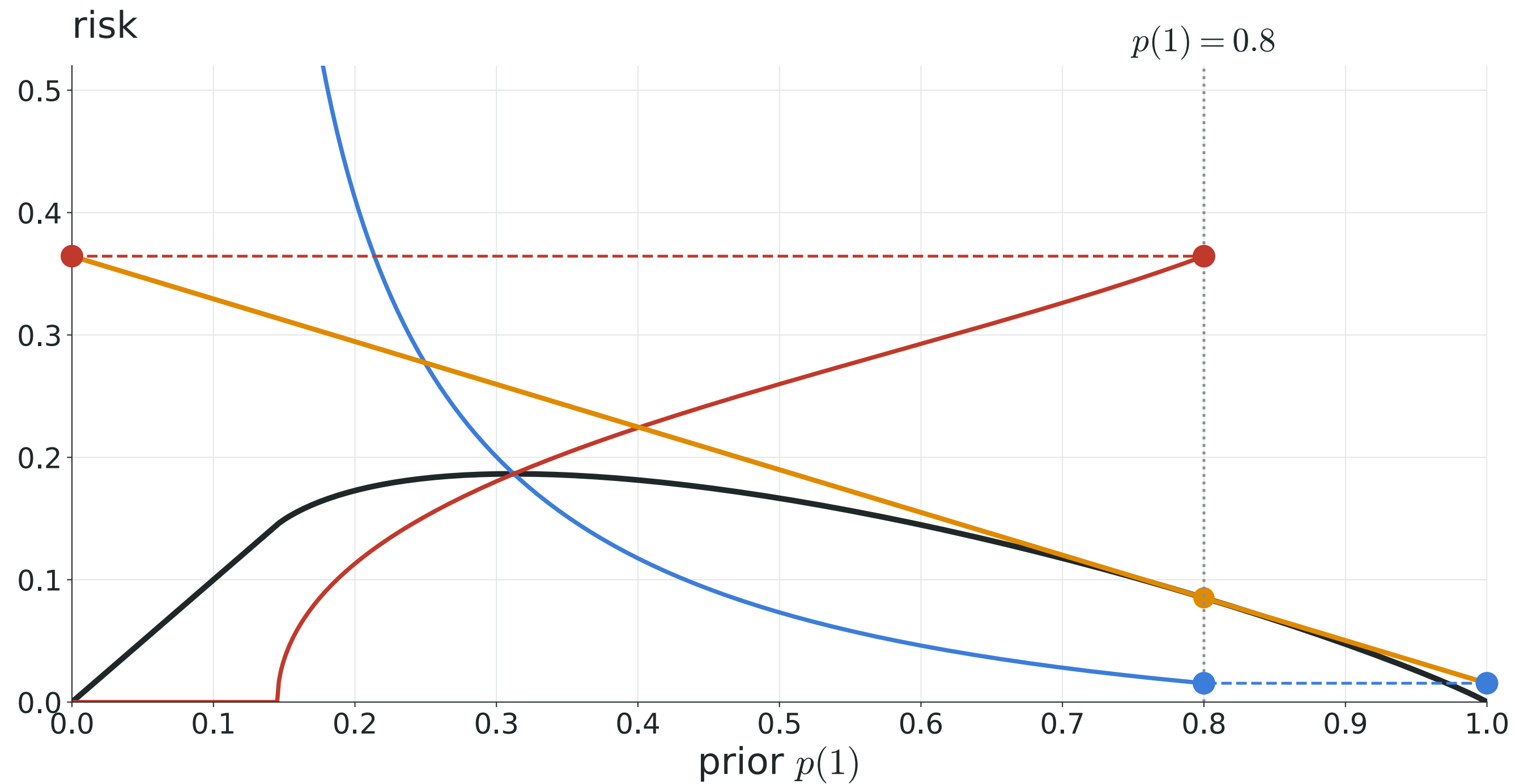
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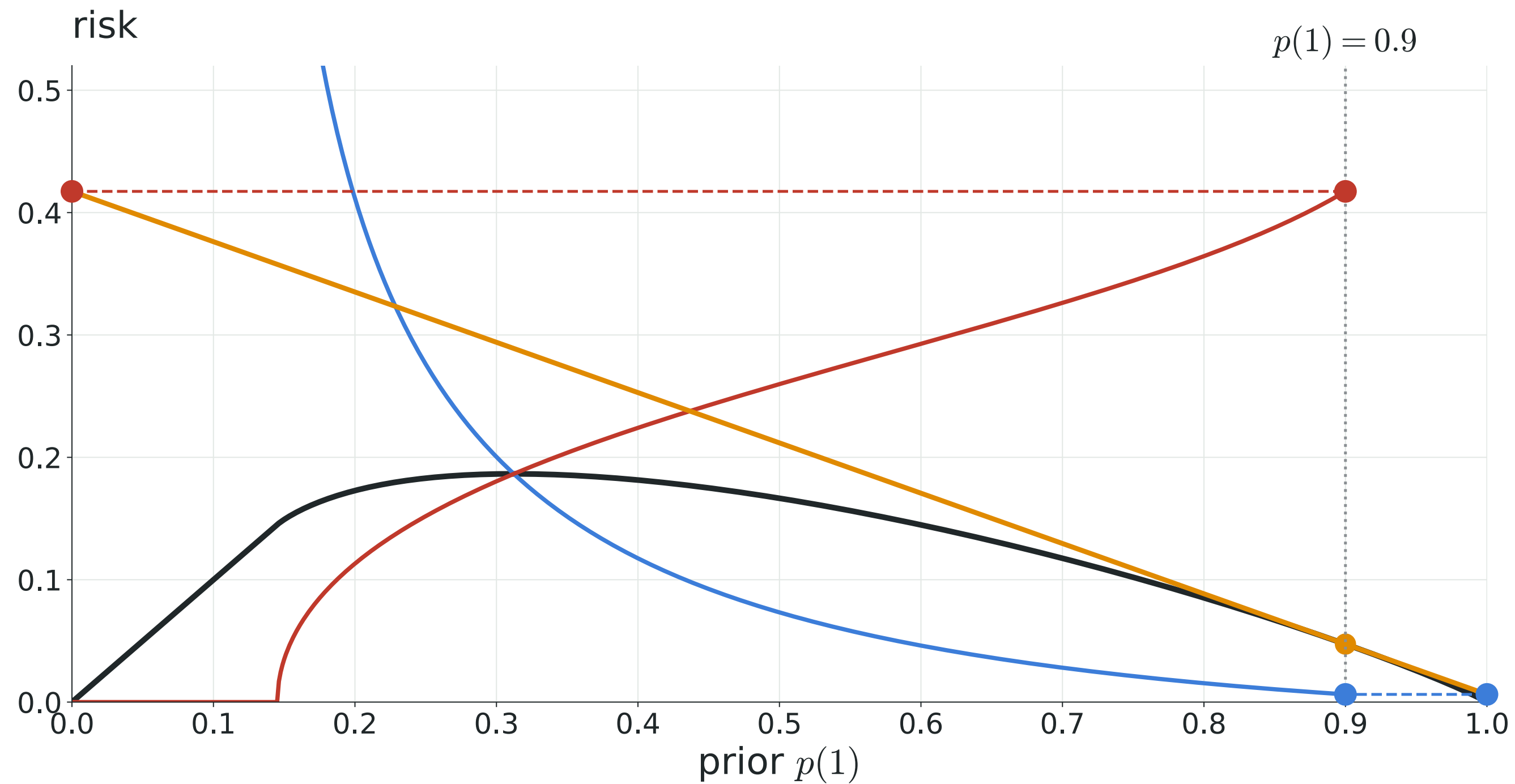
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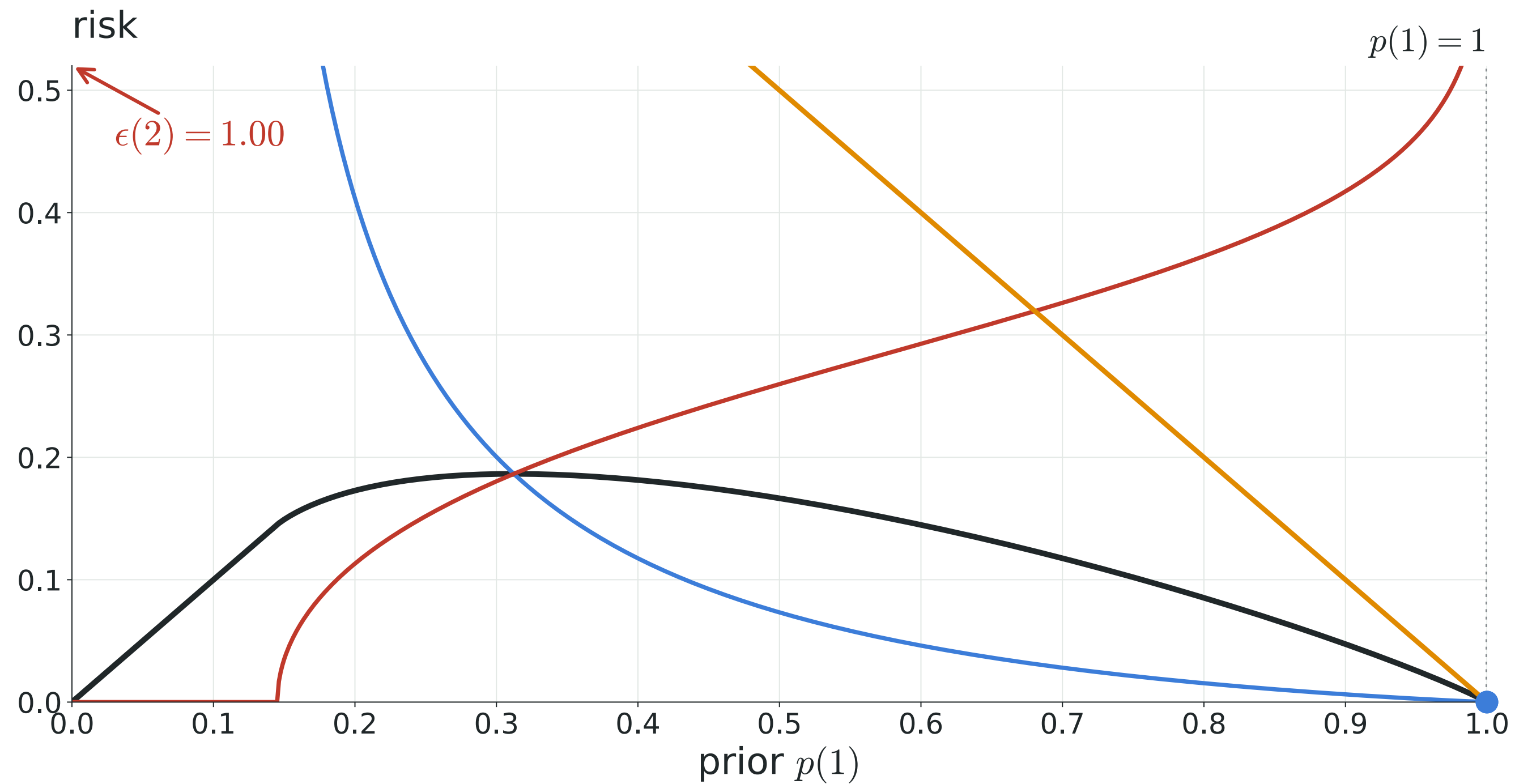
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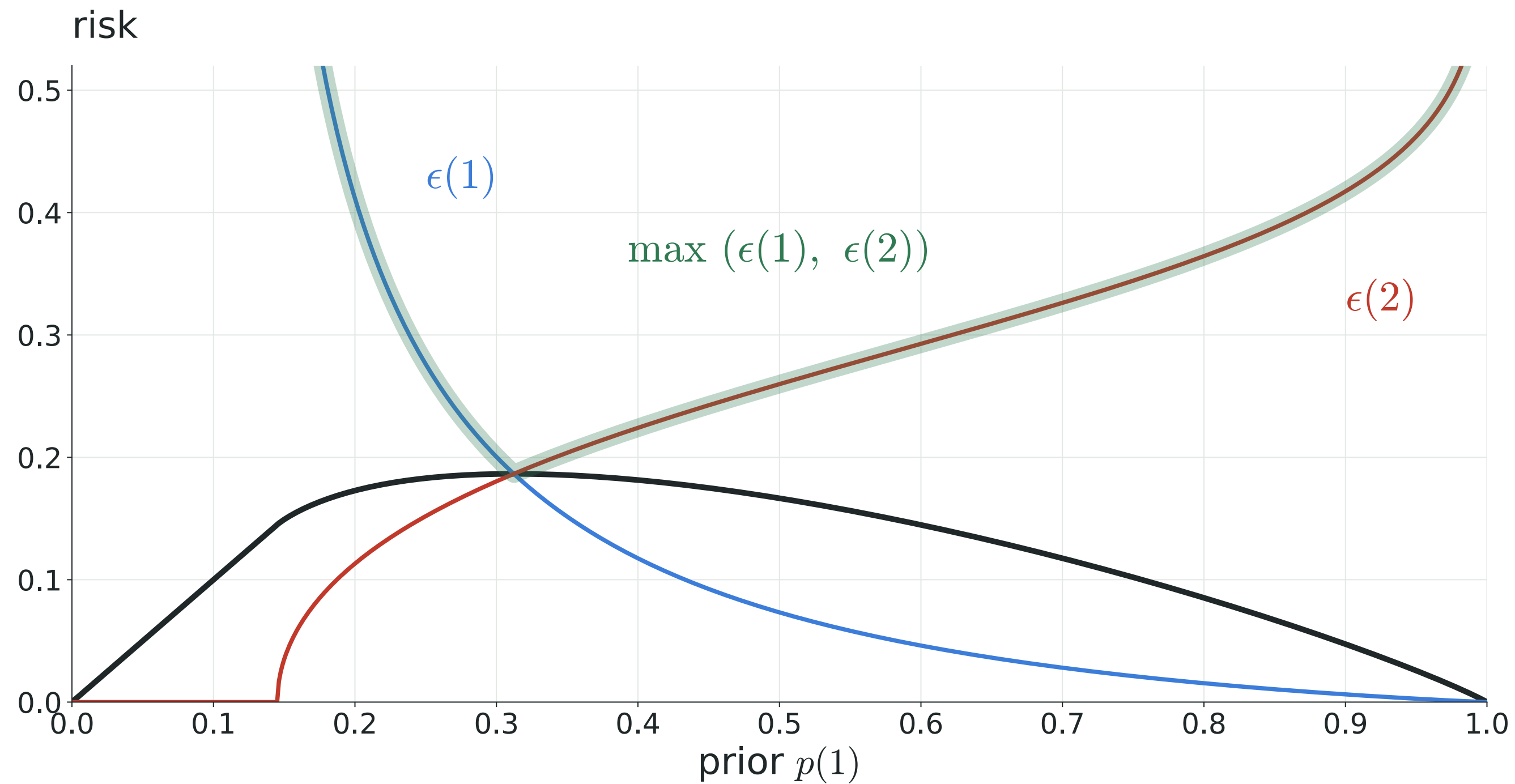
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Minimax as the worst-case Bayes



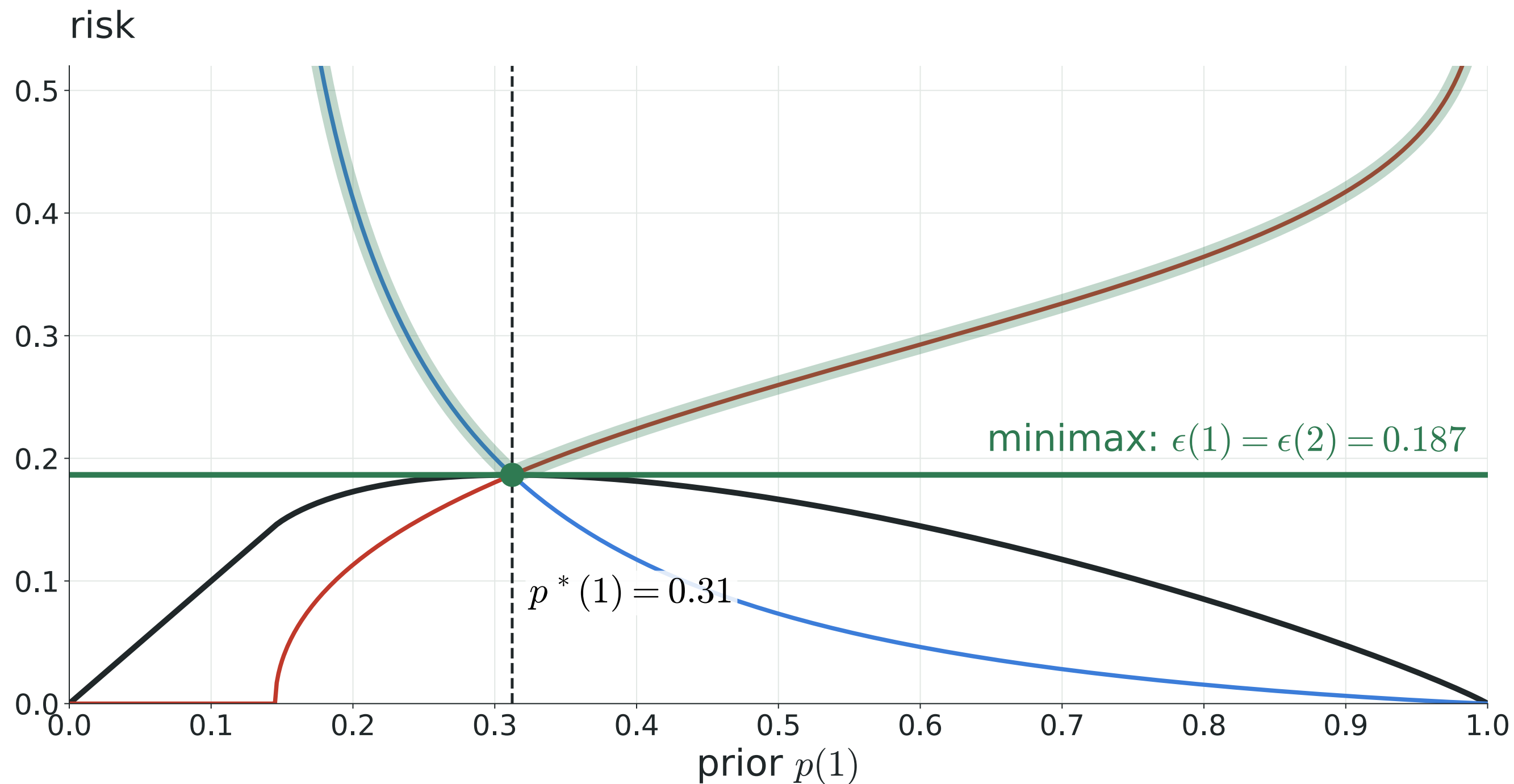
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Minimax as the worst-case Bayes



The worst case of every Bayesian strategy: $\max(\epsilon(1), \epsilon(2))$

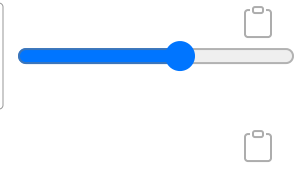
Minimax as the worst-case Bayes



Minimax: the Bayesian strategy for $p^*(1)$, its line is horizontal, $\epsilon(1) = \epsilon(2)$

Minimax and Bayes: try it

$p(1)$



↑ risk

$$\lambda = 0.67$$

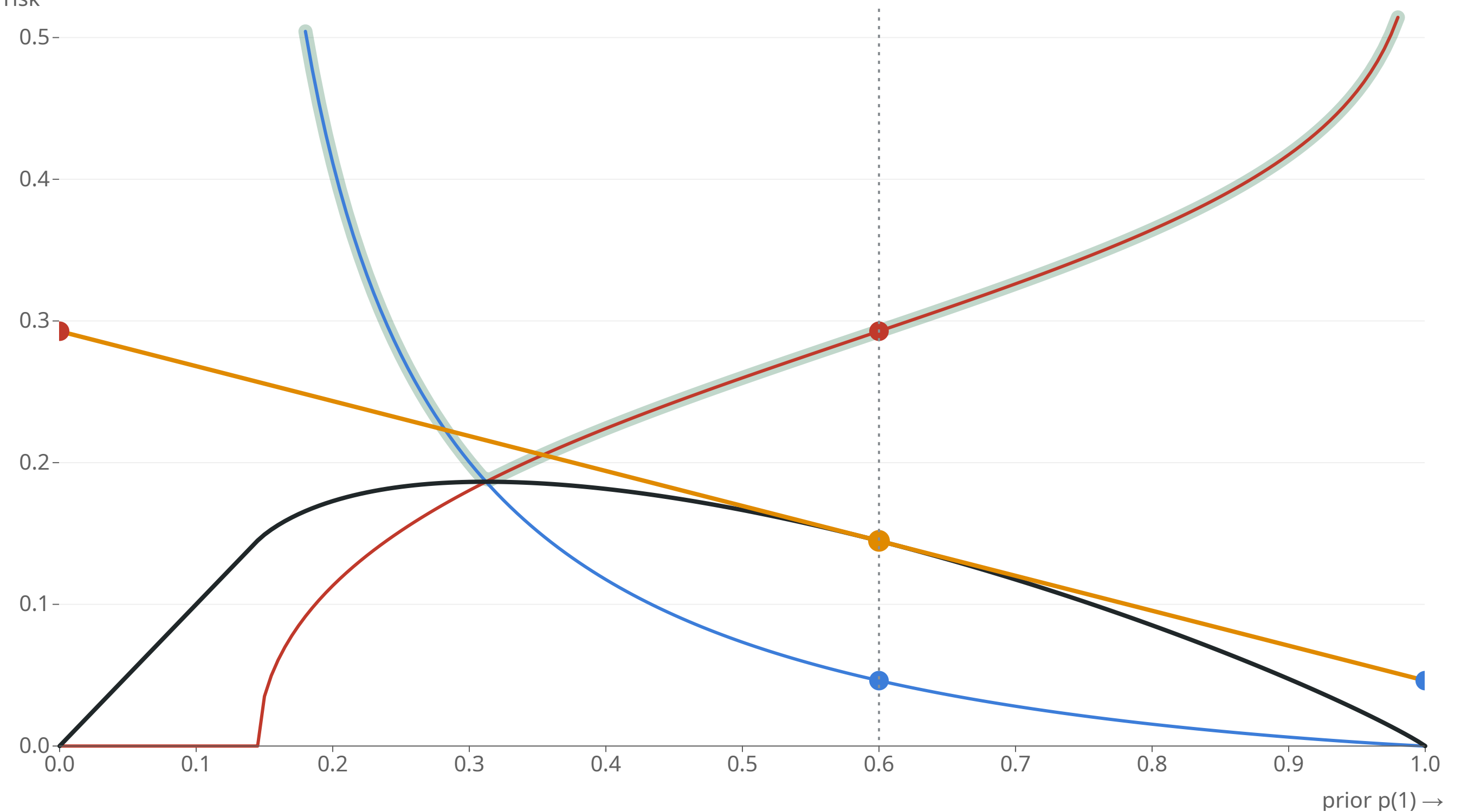
$$t_1 = -10.70$$

$$t_2 = -1.59$$

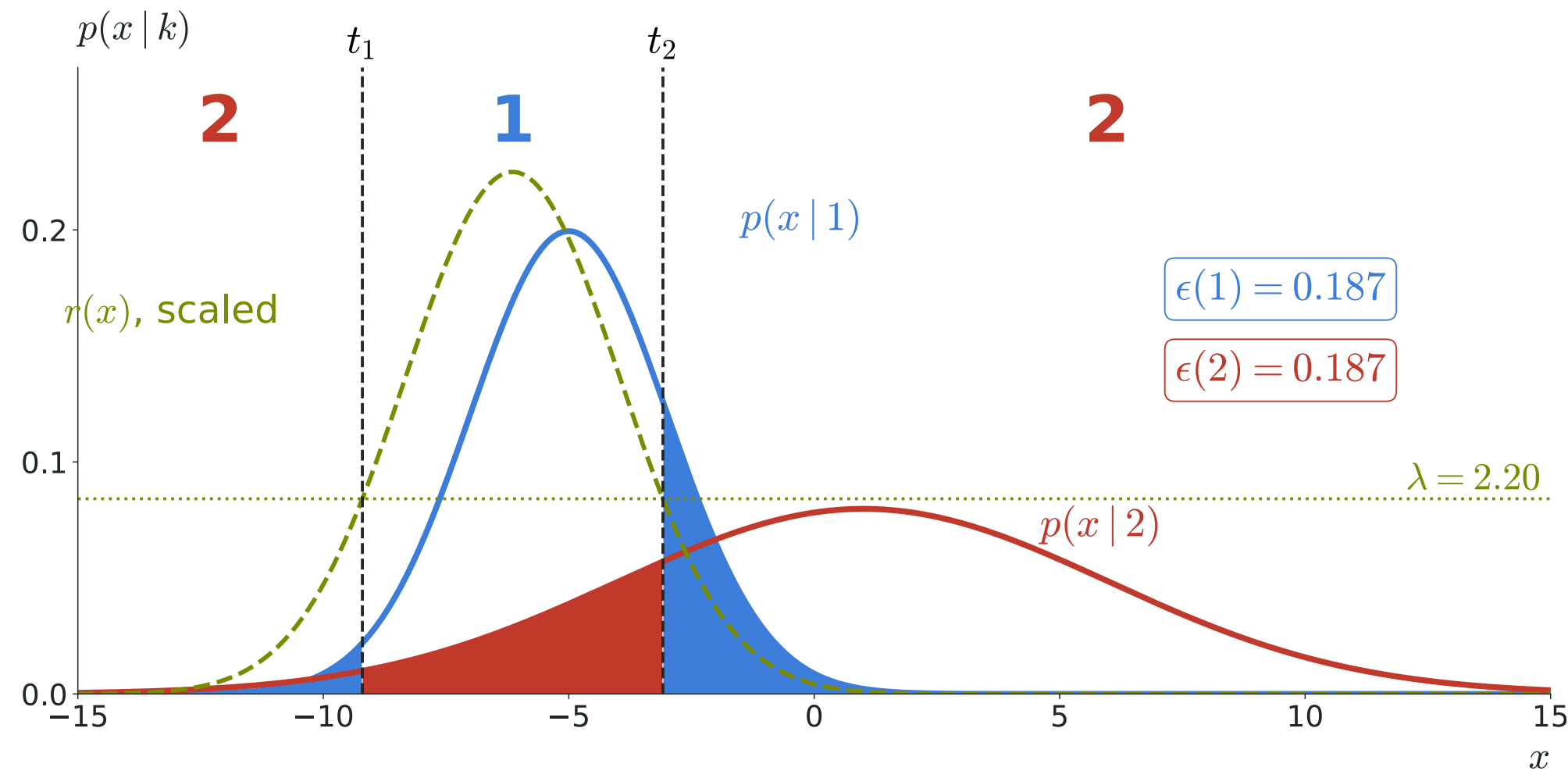
$$\varepsilon(1) = 0.046$$

$$\varepsilon(2) = 0.293$$

$$\max = 0.293$$



The minimax strategy for example III



Search over the prior instead of λ

```
1 from scipy.optimize import fminbound
2 p1 = fminbound(worst_risk, 0, 1)
```

$$p^*(1) = 0.312, \quad \lambda^* = 2.20$$

$$t_1 = -9.20, \quad t_2 = -3.08$$

$$\epsilon(1) = \epsilon(2) = 0.187$$

$$q^*(x) = \begin{cases} 1 & \text{if } t_1 < x < t_2 \\ 2 & \text{otherwise} \end{cases}$$



A grid of priors is **too coarse**.

Output of the assignment

Part 1: discrete

- `minimax_strategy_discrete`: strategy, i^* and the errors for all $N + 1$ thresholds
- `classify_discrete`: labels for test images

Part 2: continuous

- `risk_fix_q_cont`: fixed strategy, all priors
- `worst_risk_cont`: worst risk for one prior
- `minimax_strategy_cont`: the result

! Copy your Lab 2 functions into `minimax.py`. Imports do not work in the BRUTE evaluation.

Take-away for the lab

- ✓ **Minimax** needs only $p(x \mid k)$: no priors, no loss
- ✓ The strategy is a **threshold on the likelihood ratio** $r(x)$
- ✓ **Discrete x** : sort by r , try all $N + 1$ thresholds, take the smallest maximum
- ✓ **Continuous x** : the optimum is where $\epsilon(1) = \epsilon(2)$
- ✓ **Normal distributions**: minimax is the Bayesian strategy for the **worst prior**
- ✓ **Next week: test!** Bring a calculator

